German Internal Migration: A Marriage Market Perspective *

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Abstract

This paper investigates whether the large young East–West migration after German reunification altered marriage formation and marital stability in receiving West German districts. I construct a newly assembled district panel of marriage, divorce and birth outcomes over 1985–2007 with observed cumulative young-adult arrivals from 1991 onward, scaled by fixed pre-reunification population. To address endogenous destination choice, I construct a shift-share instrument by multiplying each district's 1961 East-linked settlement share by the cumulative rate of young-adult arrivals from East Germany, calculated across sample districts in the other West German states. The preferred IV estimates show that districts receiving one additional young arrival per 1,000 residents had 0.027 fewer annual divorces per 1,000 residents, about 9 percent of the mean per standard deviation of exposure. The corresponding marriage estimate is −0.021, and statistically imprecise. Event studies on balanced district panels extending to the early 1980s do not reject the null of no differential pre-reunification divorce trends. Districts with stronger historical ties did not experience faster income growth after reunification, and the estimate is robust to controlling for income. Filing-side analyses do not detect a statistically significant difference between wife- and husband-initiated divorce responses. The evidence points to lower local divorce rates in more exposed districts.

Keywords: Migration, divorce, sex-ratio, Germany.

JEL Codes: J12, J61, R23, D10.

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1 Introduction

German reunification triggered a large and persistent wave of internal migration from East to West Germany that reshaped local demographic and economic conditions across West German districts (*Kreise* and *kreisfreie Städte*) (Fuchs-Schündeln & Schündeln, 2009; Glorius, 2010; Stawarz et al., 2020). Since reunification, East Germany experienced a large net population loss through internal migration to West Germany, exceeding 1.2 million people over the post-reunification period; gross East–West mobility was substantially larger (Rosenbaum-Feldbrügge et al., 2022; Stawarz et al., 2020). These flows were also strongly selective. Young and relatively educated women moved disproportionately, while return migration was more common at older ages (Fuchs-Schündeln & Schündeln, 2009; Kröhnert & Vollmer, 2012; Leibert, 2016). As a result, post-reunification migration changed local population composition in both origin and destination regions and contributed to pronounced sex-ratio imbalances in parts of East Germany (Gulczyński, 2023; Stawarz et al., 2024). This demographic reallocation matters because marriage and divorce affect household resources and the incentives for marriage-specific investments, including children and within-household specialization (Becker et al., 1977; Bröckel & Andreß, 2015).

This paper therefore asks: did greater exposure to young East–West migration after German reunification affect marriage formation and dissolution in receiving West German districts? When potential partners of the opposite sex become more abundant, search frictions may fall, outside options may improve, and the incentives governing marriage and divorce may change (South & Lloyd, 1992). Existing evidence suggests that the direction of these effects is context dependent. Gender imbalances have been linked to marriage, fertility, bargaining, labor supply, and long-run gender norms in a range of settings (Grosjean & Khattar, 2019; Kesternich et al., 2020; Ogasawara & Igarashi, 2025). In Denmark, opposite-sex exposure in the workplace predicts divorce risk, suggesting that partner-market opportunities may operate through concrete interaction environments rather than only broad geographic sex ratios (Uggla & Andersson, 2018). German micro-level evidence has examined district sex ratios as contextual predictors of relationship dissolution, but the estimated district-level sex-ratio effects are not robustly significant. This makes the German case an open setting rather than a settled empirical fact (Obersneider et al., 2019). Evidence from college dating markets shows that local sex ratios affect relationship behavior: where women are relatively more numerous, they are less likely to form committed relationships and more likely to report less traditional dating patterns (Uecker & Regnerus, 2010). Against this background, female-skewed East–West migration among young adults after 1990 offers a particularly informative within-country setting in which to study whether a large and demographically distinctive migration episode to receiving areas altered local marriage markets and, in turn, divorce

rates.

The direction of the effect is not immediately apparent a priori, and the joint response of the two outcomes serves to differentiate between the mechanisms involved. Arrivals affect the local pool of potential partners in two ways: more singles generate more matches (Becker et al., 1977), and a larger pool of unmarried alternatives raises the expected gains from leaving an existing union (South & Lloyd, 1995). If the female skew of the inflow matters, it weakens women's outside options relative to men's, which should lower wife-initiated relative to husband-initiated divorces (Angrist, 2002; Chiappori et al., 2002); because both sexes move and the skew is moderate, however, the induced change in local sex ratios per arrival is small. If arrivals raise local housing costs (Saiz, 2007), forming and splitting households both become more expensive, so marriages and divorces should both fall. Given that district inflows are not consistently observed jointly by age and sex over the full sample period, the measurement of exposure is based on total young arrivals throughout, and the estimates do not identify a female-specific effect.

The central empirical problem is that migrant destination choice is not random. East German migrants may have sorted into economically dynamic, socially distinctive, or geographically advantaged West German districts, and those same districts may have experienced different family trajectories even in the absence of migration. To address this endogeneity, a shift-share instrument is utilised in the spirit of Card (2001). The shares represent a historical East-linked settlement share, denoting the proportion of each district's residents in 1961 who had relocated from East Germany prior to the construction of the Berlin Wall. The shift measures cumulative young East–West arrivals to sample districts in other West German states from 1991 onward per 1,000 East-linked residents recorded in those districts in 1961. The instrument thus predicts a district's exposure by distributing the aggregate inflow across districts in proportion to their historical ties to East Germany; Section 4 details its construction. The share component follows the logic of Burchardi and Hassan (2013), who use historically rooted East-linked settlement patterns as a proxy for social ties to East Germany. This approach is motivated by evidence that such ties shaped migrant destination choice after reunification (Dorner et al., 2016). Because all districts in a state face the same shift in a given year, identification rests on the historical shares: conditional on state-by-year fixed effects and predetermined characteristics, districts with stronger historical East ties must not have followed different marriage and divorce paths for reasons other than migration (Goldsmith-Pinkham et al., 2020). Because the shares are measured in the census of June 1961, shortly before the Berlin Wall halted East–West migration for 28 years, they are not contaminated by ongoing adjustment to recent inflows, a central weakness of past-settlement instruments (Jaeger et al., 2018). Two threats remain. First, regions with stronger East ties grew faster after reunification (Burchardi & Hassan, 2013). Second, districts near the former border, which had higher 1961 shares, gained market access and lost place-based support

(Ehrlich & Seidel, 2018; Redding & Sturm, 2008). I assess the identifying assumption with event studies that use newly digitized divorce counts back to 1980, with distance-to-border-by-year fixed effects, and with newly assembled district income data for 1989–2007.

The main finding is that greater exposure to young East–West arrivals lowered divorce. Comparing the post-reunification period with 1985–1988, districts in the highest quintile of the 1961 East-linked share experienced a relative decline of approximately 0.35 annual divorces per 1,000 baseline residents compared with districts in the lowest quintile. The decline becomes larger with each successive quintile, and event studies on balanced panels reaching back to 1980 and 1982 show no differential trends before reunification. The preferred IV specification, which includes district and state-by-year fixed effects, implies that a district receiving one additional young arrival per 1,000 residents had 0.027 fewer annual divorces per 1,000 residents; a one-standard-deviation increase in exposure corresponds to about 9 percent of the mean. The first stage is strong, and weak-instrument-robust confidence sets exclude zero. The application of conditioning on pre-reunification divorce levels as opposed to district fixed effects results in a somewhat larger estimate of 0.036. The estimate is similar or larger when shocks that vary with distance to the former inner-German border are absorbed, when district income is controlled for, when the shift is constructed differently, when extreme-exposure districts are excluded, and when districts are weighted by population. Marriages do not rise in any specification. They fall in the specifications without district fixed effects, but not robustly: with district fixed effects, the decline is small and imprecise.

The response's timing and composition limit its interpretation. The decline emerged in the mid-1990s, following the largest inflow wave, and did not intensify in proportion to later arrivals. Consequently, the estimates captured a post-reunification shift that increased with predicted exposure across districts, rather than a constant effect of each additional arrival. The decline does not differ between husband- and wife-initiated divorces, and it is far too large to result from the small change in local sex ratios that a moderately female-skewed inflow implies. Together with the absence of any increase in marriage, these results are inconsistent with a thicker partner market and do not support a sex-ratio channel. The absence of any increase in marriage, as indicated by these results, is inconsistent with a thicker partner market and with a sex-ratio channel. They are, however, consistent with higher costs of forming and dissolving households or with changes in the composition of the resident population, which the district data cannot separate. The primary limitation is that, given the response's nature as a shift, identification is contingent on the absence of other early-1990s shocks that varied with historical East ties. Of the two threats identified above, border-distance-by-year fixed effects address the second, and newly assembled district income data address the first: districts with stronger historical ties did not experience the faster income growth documented by Burchardi and Hassan (2013), and the estimate is robust to controlling for income. The pre-reunification

event studies, however, have limited power against slow-moving differential trends.

This paper contributes to three strands of the literature. First, it makes a data contribution: I assemble, to my knowledge for the first time, a harmonized district-level panel of divorce, marriage and birth counts for West Germany that covers 320 districts from 1985 to 2007 and reaches back to the early 1980s for 241 districts in six states, digitized from printed publications of the state statistical offices, mapped to a stable 1991 district geography, and linked to annual district marriage counts compiled from official statistical publications. Second, the study makes a significant contribution to the extant literature on German internal migration after reunification. It demonstrates that the consequences of East-West mobility extend beyond labour-market adjustment and demographic redistribution, to include marital dissolution in the regions to which migrants are relocated (Fuchs-Schündeln & Schündeln, 2009; Glorius, 2010; Stawarz et al., 2020). Third, it contributes to the economics of marriage markets and to the literature on migration shocks and predetermined networks by showing that a large internal migration of young adults was followed by a decline in divorce rates and an absence of an increase in marriage rates in more exposed destination markets. This pattern is not consistent with the prediction that an increased number of potential partners will result in an increase in both marriage and divorce (Becker et al., 1977; South & Lloyd, 1995), and the divorce decline is far too large to operate through the induced change in local sex ratios (Angrist, 2002; Chiappori et al., 2002). The design combines the new panel with a shift-share instrument based on settlement shares recorded before the Berlin Wall was built, and tests for pre-reunification trends with the newly digitized data from the 1980s (Goldsmith-Pinkham et al., 2020; Jaeger et al., 2018).

The remainder of the paper is organized as follows. Section 2 reviews the historical background and develops the paper's testable hypotheses. Section 3 describes the data, sample construction, and descriptive evidence. Section 4 presents the empirical framework and identification strategy. Section 5 reports the main results, heterogeneity analyses, and mechanism evidence. Section 6 concludes.

2 Background

German reunification triggered one of the largest internal migration episodes in an advanced economy. The 18–24 age group accounted for roughly one third of net East–West migration losses between 1991 and 2006 (Glorius, 2010). Using official migration statistics for 1991–2012, Stauder (2018) shows that female overrepresentation in East–West migration is concentrated among the youngest adult movers. Geissler, Leopold and Pink report that among East German "home leavers" between 2000 and 2010, 24.1% of young women but only 12.7% of young men relocated to the West (Geissler et al., 2013). The demographic imprint is substantial: in the 18–29 age group the eastern sex ratio fell

to around 89 women per 100 men overall and to 80 women per 100 men in some rural districts, and recent work documents rural eastern counties with fewer than 70 women per 100 men aged 18–30 (Kröhnert & Vollmer, 2012; Stawarz et al., 2024).

SOEP-based projections for 1991–2012 indicate that, by age 28, around 87,000 East German men and 239,000 East German women had moved West in connection with education. Hence, the number of female educational migrants is nearly three times the number of male ones, with the gap being largest among highly educated women (Stauder, 2018). Micro-data further show that employed East–West migrants were less likely to be married and positively selected: in social insurance records only about 40% of employed East–West migrants in the mid-1990s were married compared to roughly 57% of East German non-migrants. Panel evidence indicates that individuals in partnerships are significantly less likely to move, while older single migrants are more likely to return to the East than to remain in the West (Brücker & Trübswetter, 2007; Fuchs-Schündeln & Schündeln, 2009). Overall, post-reunification East–West migration involved substantial flows of young adults of both sexes, with female overrepresentation particularly evident among education-related movers.

Figure 1 plots net East–West migration by sex, defined as moves from East Germany to West Germany minus moves in the opposite direction.[1] Two features stand out. First, before the opening of the Berlin Wall, recorded East–West mobility was very limited, reflecting the institutional constraints of German division rather than a normal low-migration equilibrium. Second, the opening of the Wall on 9 November 1989 and the subsequent institutional transition culminating in formal unification on 3 October 1990 coincide with an extraordinary and short-lived surge in net migration.

In the late 1980s, net internal migration fluctuates around only a few tens of thousands per year (roughly 10,000 to 30,000), with fairly similar contributions from men and women. A sharp discontinuity occurs in 1989: the net balance jumps to the order of 300,000 and remains similarly high in 1990. The 1989 annual total includes migration both before and after the opening of the Wall on 9 November and therefore captures the broader political transition, including the substantial departures that preceded the border opening. By 1991, the net balance falls markedly (to roughly 150,000), indicating fast normalization relative to the immediate shock.

By sex, the 1989–1990 surge is pronounced for both groups. Men contribute somewhat more at the peak (on the order of roughly 175,000 net) than women (roughly 130,000 to 140,000), implying that the initial wave is broad-based rather than purely female-driven. After 1991, net migration declines sharply: it drops below 100,000 by 1992 and settles

[1]Series are first aggregated at the state×year level and then summed to national totals. This figure restricts the sample to *German nationals*. For historical background and dates (Berlin Wall opening: 9 Nov 1989; formal unification: 3 Oct 1990; Two-Plus-Four Treaty entry into force: 15 Mar 1991), see, e.g., the chronology in the Federal Agency for Civic Education (bpb) and the Bundestag research note on the Two-Plus-Four Treaty.

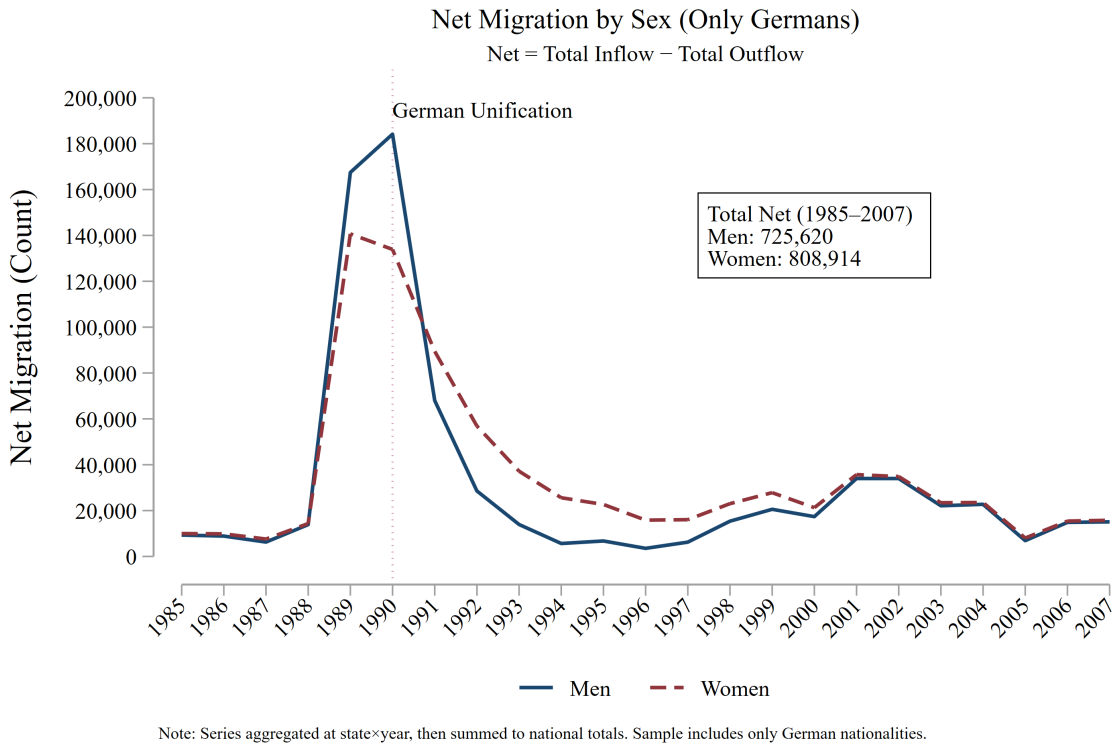


Figure 1: Net East–West internal migration by sex (German nationals only), 1985–2007

Notes: Net migration is defined as inflows from East Germany to West Germany minus outflows from West Germany to East Germany. Series are aggregated at the state×year level and summed to national East–West totals. The vertical line marks German unification.

into a much lower positive range (often 20,000 to 50,000) through much of the mid-1990s and mid-2000s. The series exhibits a secondary rise in the early 2000s, reaching roughly 70,000 at its local maximum, before returning toward the lower range.

Compositionally, the post-1991 period is more female-skewed: women account for a larger share of the net balance in many years, with male net migration reaching low levels in the mid-1990s while female net migration remains in the tens of thousands. From 1985 to 2007, the cumulative net balance is roughly 1.5 million, with women contributing slightly more than half. These net balances describe the historical migration episode; the empirical exposure measure uses cumulative gross arrivals of young adults of both sexes observed from 1991 onward, scaled by a fixed pre-reunification population.

Figure 2 shows gross arrivals from East Germany by age group. Age-specific counts start in 1991 (Section 3.2), so the 1989–1990 surge in Figure 1 cannot be broken down by age. Ages 18–29, the group whose arrivals define exposure in this paper, form the largest group in every year. Their annual number fell from 101,000 in 1991 to 44,000 in 1994, rose to a second peak of 82,000 in 2001, and stood at 67,000 in 2007; this path is the common component of the shift in the instrument (Section 4.2). Arrivals of minors fell more steeply, so the inflow became increasingly concentrated among young adults, and arrivals aged 50 and older remained small throughout.

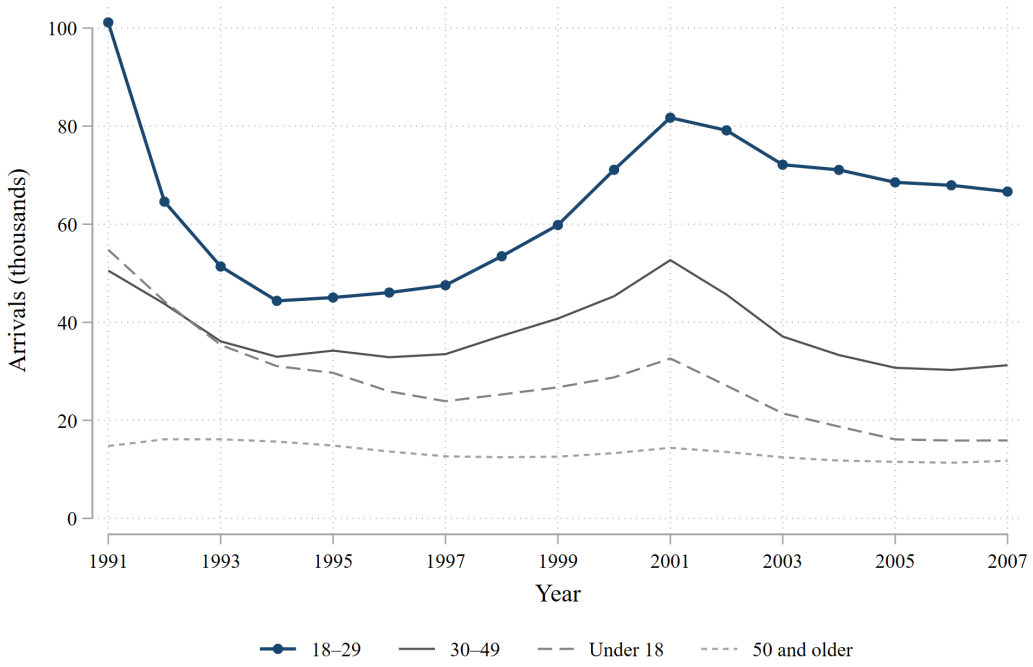


Figure 2: Arrivals from East Germany by age group, 1991–2007

Notes: Arrivals from the five eastern states (Berlin excluded) in West German districts outside the city-states, all nationalities, in thousands. Ages 18–29 as in the exposure measure (Section 3.2).

Second, the time profile mirrors an "early drop–late rebound–later decline" dynamic. From 1991 into the mid-1990s, inflows fall sharply across all major age groups (e.g., ages 18–25 drop to roughly 30,000, and under-18 inflows to roughly 25,000). From the late 1990s into the early 2000s, inflows rise again, with the rebound concentrated in the same young and prime-age groups (around 2001, ages 18–25 and 30–50 return to roughly 50,000 each, while under-18 inflows recover to roughly 30,000). After 2002, inflows decline again, especially for minors and prime-age adults, while ages 18–25 remain the single largest group through the end of the sample. Overall, the age profile supports interpreting post-1990 East-West mobility as an economically and demographically salient migration process dominated by young adults and prime-age movers, with minors moving alongside them and very limited participation of older cohorts.

Existing work on German reunification shows that historically rooted East–West ties shaped post-1989 economic behavior and spatial allocation. Burchardi and Hassan (2013) show that West German regions with stronger social ties to East Germany experienced faster post-reunification income growth, using historically determined East-linked settlement patterns and wartime destruction as a source of exogenous variation. Dorner et al. (2016) extend this logic to the migration of East German inventors and show that regions with stronger social ties to the East attracted more inventors after 1990. This literature makes it plausible that historically mediated East-links affected where migrants went after reunification. At the same time, spatial frictions and border shocks also mattered

for regional adjustment near the former inner-German border (Redding & Sturm, 2008), and place-based interventions may have had persistent local effects (Ehrlich & Seidel, 2018). These considerations motivate the historical settlement component of the instrument, while also highlighting potential violations of the exclusion restriction through other economic consequences of East–West ties and proximity to the former border.

2.1 Conceptual framework and hypotheses

I organize the possible effects of young in-migration on receiving districts with a standard model in which marriages form and dissolve according to their surplus (Becker et al., 1977; Peters, 1986). A single woman i and a single man j who meet marry, and an existing couple divorces, if

$$\text{marriage:} \quad V_{ij}^{M} - k^{F} > V_{i}^{S}(\mathcal{P}) + V_{j}^{S}(\mathcal{P}), \tag{1}$$

$$\text{divorce:} \quad V_{i}^{S}(\mathcal{P}) + V_{j}^{S}(\mathcal{P}) - k^{D} > V_{ij}^{M}, \tag{2}$$

where V_{ij}^{M} is the value of the match, which the partners can share; $V_{i}^{S}(\mathcal{P})$ is the value of being single for partner i given the local pool $\mathcal{P}$ of potential partners; and k^{F} and k^{D} are the costs of forming and of dissolving a joint household, such as the cost of one rather than two dwellings. The number of marriages also rises with the number of meetings, which increases with the number of singles in $\mathcal{P}$. Because the partners can share the surplus, whether a marriage dissolves depends only on the sum $V_{i}^{S} + V_{j}^{S}$ and not on which spouse files (Becker et al., 1977; Peters, 1986). Shocks that move the outside options of the two spouses in opposite directions primarily change the division of the surplus (Chiappori et al., 2002); they affect dissolution only to the extent that the surplus cannot be fully reallocated, which would show up as a change in which spouse initiates divorce. Young East–West migration can enter equations (1) and (2) through three channels, which generate distinct predictions for marriages, divorces, and the filing composition (Table 1).

The first channel is a thicker partner market. Young arrivals add singles to the local pool, which raises the number of meetings and the value of being single for spouses who have access to this pool (South & Lloyd, 1995).

Hypothesis 1 (thicker partner market). Exposure to young arrivals raises both marriages and divorces in receiving districts.

The second channel is the sex composition of the inflow. Among young adults, more women than men moved from East to West, so the inflow makes women relatively less scarce, lowering women's and raising men's outside options (Angrist, 2002; Chiappori et al., 2002). Because both sexes move, however, the change in the local sex ratio per arrival is small. One additional young arrival per 1,000 residents changes the ratio of men

to women aged 20–34 by approximately

$$\Delta R \approx \frac{(1-\phi) - R_0\,\phi}{F_0}, \tag{3}$$

where ϕ is the female share of young arrivals, R_0 is the baseline number of men per woman aged 20–34, and F_0 is the number of women aged 20–34 per 1,000 residents. With $\phi \approx 0.53$ at ages 18–25 (Stauder, 2018), $R_0 = 1.056$, and $F_0 \approx 115$ in the 1987 census data, $\Delta R \approx -0.0008$: even a large cumulative inflow moves local sex ratios only slightly.

Hypothesis 2 (sex composition). Exposure lowers divorce, more strongly for divorces initiated by wives, with effects bounded by the small induced change in local sex ratios.

The third channel works through the costs of forming and dissolving households. Inflows raise local housing demand and rents (Saiz, 2007), which increases both k^F and k^D: establishing a joint household and splitting one into two both become more expensive.

Hypothesis 3 (household costs). Exposure lowers both marriages and divorces, without a systematic difference between husband- and wife-initiated divorces.

Two non-behavioral mechanisms can also lower the outcomes, which are counts scaled by fixed pre-reunification population. If arrivals displace incumbent residents, as native internal migration responds to immigration in other settings (Borjas, 2006), local marriage and divorce counts fall without any change in behavior. And if fewer marriages form, fewer short-duration marriages are at risk of divorce later; because marriages that ended in divorce had lasted 13 to 14 years on average in 2000–2007 (Statistisches Bundesamt, 2025-06-26, 2025), this channel matters mainly in the later sample years.

Table 1: Predicted responses to young in-migration by channel

Channel	Marriages	Divorces	Wife- vs. husband-filed
H1: Thicker partner market	+	+	both rise
H2: Female-skewed composition	≈ 0	− (small)	wife-filed fall more
H3: Household costs	−	−	no difference
Non-behavioral mechanisms			
Displacement of residents	−	−	proportional
Fewer marriages at risk	—	− (later years)	proportional

Notes: Predicted signs for annual district counts scaled by fixed 1987 population. The divorce effect under H2 is bounded by the change in local sex ratios implied by equation (3). Under H1 and H3 the filing composition follows from the transferable-utility logic of equations (1) and (2).

The mechanism is not directly observed. I therefore use the joint response of marriages and divorces to discriminate between Hypotheses 1 and 3, and treat the filing-party decomposition, available for three states, as exploratory evidence on Hypothesis 2.

Finally, the effect may vary across local marriage markets. A given inflow should

matter more where partner markets are thinner, where migration frictions are lower, or where historical East–West ties made post-reunification mobility more salient. I therefore examine heterogeneity by district type and by distance to the former inner-German border. These tests are descriptive: they show where the response is strongest but do not by themselves establish the behavioral mechanism.

3 Data

The analysis panel covers 320 West German districts (*Kreise*), 86 independent cities and 234 rural districts, on 1991 boundaries and observed annually from 1985 to 2007.[2]

3.1 Divorces, marriages and births

Divorce counts by district come from the annual reports on court-ordered dissolutions of marriage (*Gerichtliche Ehelösungen*) that each state statistical office publishes (Statistische Landesämter, 1981–2008). I digitized the printed tables by optical character recognition. The series begin in 1980 in Baden-Württemberg, Hesse, North Rhine-Westphalia and the Saarland, in 1981 in Schleswig-Holstein, in 1982 in Bavaria, and in 1985 in Lower Saxony and Rhineland-Palatinate, for which no earlier district tables are accessible in print or in digital form. All 320 districts are therefore observed from 1985, and two balanced panels reach further back: 241 districts in six states from 1982, and 130 districts in four states from 1980.

I checked the digitized counts against the scanned tables, using the totals that each table prints for administrative regions and for the state as accounting checks. The checks found three kinds of error in an earlier transcription: values copied from an adjacent year, counts assigned to a city instead of the surrounding district of the same name or the reverse, and misread digits. Most corrections concern Bavaria (1,296 district-year cells over 1982–2007); Appendix A.1 documents the checks.

Divorces are assigned to the district of residence that determines which family court is competent (Statistisches Bundesamt, 2010c). The reports also record who filed the petition: the husband, the wife, or both spouses. Section 5.4 uses this information for Lower Saxony, North Rhine-Westphalia and Baden-Württemberg, which I digitized for 1992–2007: 141 districts, with Lower Saxony missing in 1995 and 1997.

Marriages and live births, in total and to unmarried parents, come from the Federal Statistical Office: for 1985–1999 from the annual volumes *Gebiet und Bevölkerung* (Fachserie 1, Reihe 1), which I digitized in the same way, and for 2000–2007 from the digital

[2]The sample excludes the city-states (West Berlin, which the 1961 district data do not cover, Hamburg and Bremen) and four districts of Lower Saxony that cannot be matched across all sources on constant boundaries: the city and the district of Hannover, which merged into the Region Hannover in 2001, and the districts of Göttingen and Osterode am Harz, which merged in 2016.

district table N 50, *Eheschließungen, Geborene und Gestorbene nach Kreisen* (Statistisches Bundesamt, 1987–2001, 2000–2007). Births are assigned to the mother's district of residence (Statistisches Bundesamt, 2010a). Marriages are assigned to the registry office where the wedding takes place, wherever the spouses live (Statistisches Bundesamt, 2010b). Districts that attract weddings from elsewhere therefore record more marriages than their residents enter into. The clearest cases are Nordfriesland, Friesland and Wittmund, which include North Sea islands, and the spa town of Baden-Baden; Nordfriesland records up to 16 marriages per 1,000 residents a year, against a sample mean of 6.

All outcomes are expressed per 1,000 residents counted in the census of 25 May 1987:

$$y_{it} = 1000 \, \frac{E_{it}}{P_{i,1987}}, \tag{4}$$

where E_{it} is the number of divorces, marriages or live births in district i in year t, and $P_{i,1987}$ is the district's census population (Schmitt et al., 1994). Because the denominator is fixed, arrivals cannot lower the rates mechanically by enlarging the population. y_{it} is therefore a count scaled by a constant, not a hazard among married couples.

3.2 Migration from East Germany

Arrivals come from the annual district-to-district migration matrices of the statistical offices (*Kreiswanderungsmatrix*), which are available in digital form from 1991 (Statistisches Bundesamt, 1991–2007). Each matrix counts registered moves of a main residence between districts by district of origin, district of destination and age group; from 2000 on, it also reports sex and nationality.[3]

For each district i and year τ, $M_{i\tau}$ is the number of arrivals aged 18–29 (the matrix's age groups 18–24 and 25–29) from districts in Brandenburg, Mecklenburg-Western Pomerania, Saxony, Saxony-Anhalt and Thuringia; moves from Berlin are not counted. Exposure is the cumulative inflow up to the previous year per 1,000 residents:

$$X_{it} = \frac{1000}{P_{i,1988}} \sum_{\tau=1991}^{t-1} M_{i\tau}, \tag{5}$$

where $P_{i,1988}$ is the district's population on 31 December 1988, the last year-end before the border opened (Statistisches Bundesamt, 1989), and $X_{it} = 0$ for $t \leq 1991$. X_{it} counts gross arrivals: neither return moves to the East nor onward moves within the West are netted out. Because the matrices begin in 1991, arrivals in 1989 and 1990 are not observed at the district level.

[3]The unit is a move, not a person: someone who moves twice in a year is counted twice. Age refers to the end of the reporting year. Moves of foreign nationals are included.

The matrices do not report sex before 2000. The printed migration reports of the state statistical offices give the sex composition of arrivals from the eastern states at the state level (Statistische Landesämter, 1992–2008); I digitized them, and they enter only the calculation in Section 2.1.

3.3 The 1961 East-linked share

The census of 6 June 1961 recorded as a separate group the Germans who had lived in the Soviet zone or East Berlin on 1 September 1939 and moved to the West after the war, together with their children born since; expellees from the former eastern territories were counted separately (Ritter & Niehuss, 1987, p. 31). District counts of this group, R_i, come from the GESIS edition of the 1961 census district data (Blasius & Antoine, 1990), converted from the 565 districts of 1961 to 1991 boundaries ((Appendix A.3)). The share is

$$s_i = \frac{R_i}{P_{i,1961}}, \tag{6}$$

where $P_{i,1961}$ is the district's 1961 population on the same boundaries. In the seven states whose districts are all in the sample, the district counts summed to the state reproduce the published census shares, for example 3.1 percent in Bavaria, 5.7 percent in North Rhine-Westphalia and 6.3 percent in Hesse (Ritter & Niehuss, 1987, p. 31); Appendix Table A.1 reports all states.

3.4 Baseline controls

The controls are district characteristics measured before reunification, all from the GESIS district file of the 1950–1987 censuses (Schmitt et al., 1994). From the 1987 census, they are the share never married, log population density, the in-commuting rate, the unemployment rate, and the population share and the sex ratio (men per woman) at ages 20–34. From the 1970 census, they are the Catholic share of the population and women's share of the employed. The robustness and heterogeneity analyses also use each district's distance to the former inner-German border, computed from the federal administrative boundary files (Bundesamt für Kartographie und Geodäsie, 1998). Table 2 reports descriptive statistics.

4 Identification Strategy

This section sets out the estimating equations (Section 4.1), the instrument (Section 4.2), and the identifying assumption together with the threats to it (Section 4.3).

Table 2: Descriptive statistics

	Mean	SD	P10	P90
A. Outcomes per 1,000 residents of 1987, 1992–2007				
Divorces	2.52	0.53	1.86	3.17
Marriages	6.02	1.05	4.81	7.18
Live births	10.73	1.74	8.53	13.00
B. Migration and settlement				
Exposure, 1992–2007	9.66	7.93	2.33	19.55
Exposure in 2007	18.41	10.47	7.42	32.65
1961 East-linked share (%)	4.28	2.01	1.92	7.07
C. Baseline controls				
Share never married, 1987 (%)	38.61	2.59	35.40	41.70
Log population density, 1987	5.63	1.08	4.43	7.28
In-commuting rate, 1987 (%)	17.01	7.91	9.92	28.46
Unemployment rate, 1987 (%)	6.63	2.53	3.63	10.14
Population share aged 20–34, 1987 (%)	23.70	1.75	21.96	25.48
Sex ratio aged 20–34, 1987	1.06	0.04	1.01	1.11
Catholic share, 1970 (%)	50.11	28.69	8.68	86.53
Women's share of the employed, 1970 (%)	35.99	4.09	30.38	41.49

Notes: Divorces, marriages, live births and exposure in 1992–2007: 5,120 district-years (320 districts, 16 years). Exposure in 2007, the share and the controls: 320 districts. Outcomes are defined in equation (4), exposure (cumulative arrivals aged 18–29 per 1,000 residents of 1988) in equation (5), and the share in equation (6). Share never married: residents of all ages. Sex ratio: men per woman. P10 and P90: 10th and 90th percentiles.

4.1 Estimating equations

The preferred specification is

$$y_{it} = \beta X_{it} + \alpha_i + \delta_{s(i)t} + W_i'\gamma_t + \varepsilon_{it}, \tag{7}$$

where y_{it} is the outcome (equation (4)), X_{it} is exposure (equation (5)), α_i is a district fixed effect, $\delta_{s(i)t}$ is a fixed effect for the state $s(i)$ of district i in year t, and W_i is the vector of baseline controls (Section 3.4), whose coefficients γ_t differ by year. Because arrivals chose their destinations, exposure may be correlated with ε_{it}. I therefore estimate β by two-stage least squares, with first stage and reduced form

$$X_{it} = \pi Z_{it} + \alpha_i^{\pi} + \delta_{s(i)t}^{\pi} + W_i'\gamma_t^{\pi} + u_{it}, \tag{8}$$

$$y_{it} = \rho Z_{it} + \alpha_i^{\rho} + \delta_{s(i)t}^{\rho} + W_i'\gamma_t^{\rho} + e_{it}, \tag{9}$$

where Z_{it} is the instrument defined in equation (10), and each equation has its own fixed effects and control coefficients. With one instrument and one endogenous regressor, $\hat{\beta} = \hat{\rho}/\hat{\pi}$. The coefficient β is the change in annual events per 1,000 residents of 1987

when cumulative arrivals rise by one per 1,000 residents of 1988.

Exposure ends in the year before the outcome because divorce responds with a delay: a marriage can normally be dissolved only after the spouses have lived apart for a year (§ 1565 BGB). I use the same timing for marriages and births. Measuring exposure as a cumulative stock imposes that each arrival changes the outcome by the same amount in every later year, as the mechanisms in Section 2.1 imply. Section 5.1 relaxes this restriction by estimating the effect separately for three periods.

The tables report three specifications. The first includes only state-by-year fixed effects and compares outcome levels across districts of the same state and year. The second, a lagged-outcome specification, adds the baseline controls and the outcome's 1987 level with coefficients that are constant over time; it compares districts that had the same outcome before reunification. Both use 1992–2007. The third is equation (7). Its district fixed effects remove permanent differences between districts, and its year-specific control coefficients let districts with different baseline structures follow different paths, so that urban districts, for example, can follow a different divorce path from rural ones; Appendix B.4 reports the estimates without these coefficients. All controls are measured before reunification, so none can respond to the arrivals. This specification adds 1985–1988, when exposure is zero, so that the district effects are estimated from years before any arrivals. It omits 1989–1991, when arrivals were already large but exposure is recorded as zero (Section 3.2).

The lagged-outcome and fixed-effects specifications rest on different assumptions: that districts with the same 1987 outcome are comparable, and that districts would have followed parallel paths once permanent differences are removed. In a simpler setting with a binary treatment, Ding and Li (2019) show that the two approaches bracket the true effect; I report both and read the range between them as the plausible range of the effect. I prefer the fixed-effects specification because its identifying assumption can be tested with pre-reunification data (Section 4.3) and because it removes all permanent differences between districts, not only those captured by the 1987 outcome, whose association with later divorce changes over time (Section 5.3).

Standard errors are clustered by district. Because the model is just identified, the Anderson–Rubin test of $\beta = b_0$ is the test that Z_{it} has no effect in the reduced form with $y_{it} - b_0 X_{it}$ as the outcome. The values of b_0 that this test does not reject form a confidence set that remains valid when the instrument is weak (Anderson & Rubin, 1949). I report these sets together with the Kleibergen–Paap first-stage F statistic (Kleibergen & Paap, 2006).

4.2 The instrument

The instrument predicts each district's arrivals from its historical ties and from the rate at which such ties attracted arrivals elsewhere:

$$Z_{it} = s_i \sum_{\tau=1991}^{t-1} g_{-s(i),\tau}, \qquad g_{-s,\tau} = 1000 \, \frac{\sum_{j \notin s} M_{j\tau}}{\sum_{j \notin s} R_j}, \tag{10}$$

where s_i is the 1961 East-linked share (equation (6)), $M_{j\tau}$ is the number of young arrivals in district j in year τ (Section 3.2), R_j is district j's 1961 count of East-linked residents, and the sums run over the sample districts outside state s. As with exposure, $Z_{it} = 0$ for $t \leq 1991$.

The arrival rate $g_{-s,\tau}$ measures young East–West arrivals to sample districts in other West German states per 1,000 East-linked residents recorded in those districts in 1961. Z_{it} is therefore the exposure district i would have accumulated if its arrivals had followed its ties at the rate observed elsewhere, in the spirit of instruments built on earlier immigrant settlement (Card, 2001). If arrivals had been allocated exactly in proportion to ties, the first-stage coefficient π would equal $P_{i,1961}/P_{i,1988}$, about 0.9 on average, which gives the first-stage estimate a natural scale.

For each district, the aggregate arrival rate is calculated using sample districts in other West German states. Arrivals to the district's own state therefore do not enter this component of the instrument. State-by-year fixed effects absorb shocks common to districts within the same state and year. The robustness analysis also considers rates calculated from all other sample districts, including those in the same state, and from the full sample (Section 5.2).

Within a state and year, all districts use the same cumulative arrival rate calculated from the other West German states. With state-by-year fixed effects, the instrument therefore varies across districts only through s_i. The cumulative arrival rate determines the timing and scale of the comparison, while the historical share determines which districts are compared. Identification consequently rests on the historical shares, as in Goldsmith-Pinkham et al. (2020). The eight state-specific arrival-rate series largely reflect the same aggregate migration episode and therefore do not provide the many independent shocks required for an identification argument based on shock variation (Borusyak et al., 2025).

Instruments based on past settlement can also confound the response to new arrivals with continuing adjustment to earlier inflows from the same origin (Jaeger et al., 2018). Here the inflows that formed s_i ended when the Wall closed the border in 1961, 28 years before the new inflows began. Adjustment still under way in the 1980s would appear in the pre-reunification tests of Section 5.3.

4.3 Identifying assumption and threats

Because the instrument varies within a state and year only through s_i, the reduced form (9) is a restricted version of the event study

$$y_{it} = \sum_{k \neq 1988} \theta_k \, s_i \, \mathbf{1}[t = k] + \alpha_i + \delta_{s(i)t} + W_i' \gamma_t + \varepsilon_{it}, \tag{11}$$

where s_i is measured in percentage points and θ_k is the difference in the outcome in year k, relative to 1988, per percentage point of the share. The reduced form restricts θ_k to be proportional to the cumulative arrival rate in other West German states up to year $k-1$, apart from small differences in that rate between states: zero through 1991 and increasing with cumulative arrivals thereafter.

The identifying assumption is that, without the arrivals, outcomes would have evolved alike in districts of the same state with different shares, given the controls:

$$\mathbb{E}\big[\, y_{it}(0) - y_{i,1988}(0) \,\big|\, s_i, W_i, s(i) \big] = \mathbb{E}\big[\, y_{it}(0) - y_{i,1988}(0) \,\big|\, W_i, s(i) \big], \tag{12}$$

where $y_{it}(0)$ is the outcome district i would have had in year t without arrivals from the East. Before 1989, $y_{it}(0) = y_{it}$, so the assumption implies $\theta_k = 0$ in every pre-reunification year, which Section 5.3 tests. After 1991, it is the exclusion restriction: arrivals must be the only channel through which historical ties changed outcomes. Pre-reunification data cannot test this part. The main threats are the following.

1. *Slow-moving differential trends.* Districts with stronger ties could have been on gradually diverging paths that a pre-period of four to nine years cannot detect. Section 5.3 reports how large a trend the tests could miss.

2. *Other effects of East ties.* Regions with stronger social ties to the East grew faster after reunification (Burchardi & Hassan, 2013), and ties may also have revived contact with relatives across the former border. Either could change family outcomes directly; Section 5.3 shows that the first is absent within states. Births, which arrivals should change only through their own fertility, move in line with that benchmark (Appendix B.4). This rules out a broad demographic response to ties, but not a channel specific to marriage or divorce.

3. *Shocks near the former border.* Districts close to the border gained market access after reunification (Redding & Sturm, 2008) and lost place-based subsidies when support for the former *Zonenrandgebiet* was scaled back after 1991 and ended in 1994 (Ehrlich & Seidel, 2018). To the extent that ties are stronger near the border, these shocks load on s_i. Section 5.2 adds bins of distance to the border interacted with year indicators.

4. *Unmeasured early arrivals.* If the arrivals of 1989 and 1990 also followed historical ties, the reduced form contains their effect on later outcomes while measured exposure does not, and β overstates the effect per arrival.

5. *Spillovers between districts.* Marriage markets extend beyond district borders, so arrivals can affect neighbouring districts. Spillovers into districts with lower shares bias the estimate toward zero. Correlated shocks across neighbouring districts are handled by clustering at the level of the 81 spatial planning regions (Section 5.2).

Three features of the design shape the interpretation of β. First, it is a local estimand: it averages the effects of arrivals whose destinations followed historical ties, weighting districts by how strongly their arrivals responded to those ties. Second, the outcomes count events per baseline resident (Section 3.1), so β combines changes in the rate at which couples marry or divorce with changes in the number of people at risk. If arrivals displaced married residents to other districts (Borjas, 2006), divorce counts would fall without any change in the divorce hazard. Third, the arrivals' own events enter the counts. For divorce this contribution is positive but small, because young arrivals rarely divorce, so a negative β understates the decline among couples already living in the district. For marriages and births, the arrivals' own rates provide a benchmark against which the estimates can be judged.

5 Results

I begin by assessing whether the instrument defined in (10) predicts migration exposure and by reporting the reduced-form relationship between the instrument and the two outcomes. Throughout the section, the tables report the three specifications defined in Section 4.1: state-by-year fixed effects only, the lagged-outcome specification, and the preferred specification.

Table 3 reports two first-stage exercises. Columns (1)–(3) regress annual inflow intensity, young East–West arrivals per 1,000 baseline residents in a given year, on the instrument. This is a construction check: it shows that the instrument predicts the flow of new arrivals and not only their accumulated stock. Columns (4)–(6) report the first stage for lagged cumulative exposure, the endogenous regressor in the 2SLS estimates.

The instrument predicts exposure in every specification. In the preferred specification, one additional predicted arrival per 1,000 residents in 1961 raises measured cumulative arrivals by 0.55 per 1,000 baseline residents, with a first-stage F statistic of 27.4; the corresponding coefficients in the other specifications are 0.46 and 0.51. The instrument also predicts annual inflows, with F statistics between 32 and 73. These results establish relevance, not validity. Whether historical East-linked settlement shares are unrelated to other determinants of marriage and divorce is assessed in Section 5.3.

Table 3: First-stage evidence for the instrument

	Construction check			Main first stage		
	Annual young inflow intensity			Lagged cumulative exposure		
	(1)	(2)	(3)	(4)	(5)	(6)
Instrument	0.0608***	0.0564***	0.0418***	0.4553***	0.5085***	0.5483***
	(0.0071)	(0.0077)	(0.0074)	(0.0751)	(0.0817)	(0.1048)
State×year fixed effects	✓	✓	✓	✓	✓	✓
Baseline controls		✓			✓	
District fixed effects			✓			✓
Baseline controls×year			✓			✓
Pre-Years (1985–1988)			✓			✓
Observations	5,120	5,120	6,400	5,120	5,120	6,400
R^2	0.4596	0.5082	0.8657	0.6908	0.7197	0.9122
Instrument F statistic	72.56	53.30	32.33	36.72	38.71	27.39

Notes: Columns (1)–(3) regress annual inflow intensity, arrivals from East Germany aged 18–29 in year t per 1,000 residents of the fixed 1988 population, on the instrument. Columns (4)–(6) regress lagged cumulative exposure, the endogenous regressor in the 2SLS estimates, on the instrument. The instrument is the one-year-lagged shift-share predictor described in Section 4. Columns (1), (2), (4), and (5) use 1992–2007; columns (3) and (6) add 1985–1988, when exposure and the instrument are zero by construction, and exclude 1989–1991. Columns (2) and (5) include the baseline controls (Section 3.4) and the 1987 divorce rate; columns (3) and (6) interact the baseline controls with year indicators. Standard errors clustered by district in parentheses; the F statistic is the cluster-robust test of the instrument. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 reports the reduced form. Because the model is just identified, each 2SLS estimate below equals the ratio of the reduced-form coefficient to the first-stage coefficient, so the reduced form already determines the sign and significance of the IV results.

For divorce (Panel A), the reduced form is positive with state-by-year fixed effects only and negative once permanent differences between districts are taken into account, either through the 1987 divorce level (-0.018) or through district fixed effects (-0.015). The positive coefficient in column (1) therefore reflects that districts with stronger historical East ties had persistently higher divorce levels; the event studies below show that these differences were stable before reunification. For marriage (Panel B), the reduced form is negative in all three columns, but in the preferred specification it is small and not statistically significant (-0.012, s.e. 0.011).

5.1 Main findings

Table 5 reports OLS and 2SLS estimates for divorce (Panel A) and marriage (Panel B). The dependent variables are annual divorces and marriages per 1,000 residents of the fixed 1987 population; exposure is cumulative young arrivals per 1,000 residents of the fixed 1988 population.

In the preferred specification, exposure lowers divorce. The 2SLS coefficient in col-

Table 4: Reduced-form relationship between the instrument and the outcomes

	State-year FE (1)	Lagged outcome (2)	Preferred (3)
Panel A. Divorces per 1,000 residents			
Instrument	0.0151***	−0.0182***	−0.0150***
	(0.0039)	(0.0038)	(0.0050)
Panel B. Marriages per 1,000 residents			
Instrument	−0.0482***	−0.0289***	−0.0117
	(0.0111)	(0.0104)	(0.0106)
State × year fixed effects	✓	✓	✓
Baseline controls		✓	
District fixed effects			✓
Baseline controls × year			✓
Pre-reunification years (1985–1988)			✓
Observations	5,120	5,120	6,400

Notes: The dependent variables are annual divorces (Panel A) and marriages (Panel B) per 1,000 residents of the fixed 1987 population. The regressor is the one-year-lagged shift-share instrument. Samples and controls are as in Table 3; in Panel B, column (2) uses the 1987 marriage rate in place of the 1987 divorce rate. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

umn (6) is −0.027: a district receiving one additional young arrival per 1,000 residents had 0.027 fewer annual divorces per 1,000 residents. The first stage is strong (Kleibergen–Paap $F = 27.4$), and the Anderson–Rubin confidence set, which remains valid under weak instruments, excludes zero. The standard deviation of cumulative exposure over 1992–2007 is 7.9 arrivals per 1,000 residents, so a one-standard-deviation increase corresponds to 0.22 fewer divorces per 1,000, about 9 percent of the sample mean of 2.52.

The lagged-outcome specification in column (5) yields a somewhat larger decline of 0.036, about 11 percent of the mean per standard deviation; following Section 4.1, I read the range between the two estimates as the plausible range of the effect. With state-by-year fixed effects alone (column 4), the coefficient is positive, reflecting the persistent level differences discussed above; this specification is not informative about the effect of exposure.

The OLS estimates are close to zero in all specifications, including −0.005 in the preferred one. Two forces plausibly bias OLS toward zero. Migrants are likely to have moved toward economically dynamic districts in which divorce would otherwise have risen, and exposure is measured as gross arrivals, which overstates the number of migrants who remained and attenuates OLS. The 2SLS estimate is also local: it is identified from districts whose exposure was shaped by historical East ties.

The magnitudes should be read with two qualifications. At the 2007 mean of cumulative exposure, 18.4 arrivals per 1,000 residents, the preferred estimate implies about 0.50

Table 5: Young East–West migration exposure, divorce, and marriage

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Divorces per 1,000 residents						
Cumulative exposure $_{t-1}$	0.003	−0.003	−0.005*	0.033***	−0.036***	−0.027***
	(0.003)	(0.003)	(0.003)	(0.011)	(0.008)	(0.009)
Kleibergen–Paap F				36.72	38.71	27.40
Panel B. Marriages per 1,000 residents						
Cumulative exposure $_{t-1}$	0.010	0.009	0.007	−0.106***	−0.056**	−0.021
	(0.013)	(0.009)	(0.008)	(0.031)	(0.022)	(0.020)
Kleibergen–Paap F				36.72	37.44	27.40
State × year fixed effects	✓	✓	✓	✓	✓	✓
Baseline controls		✓			✓	
District fixed effects			✓			✓
Baseline controls × year			✓			✓
Pre-Years (1985–1988)			✓			✓
Observations	5,120	5,120	6,400	5,120	5,120	6,400

Notes: The dependent variables are annual divorces (Panel A) and marriages (Panel B) per 1,000 residents of the fixed 1987 population. Exposure is cumulative arrivals from East Germany aged 18–29 between 1991 and $t-1$ per 1,000 residents of the fixed 1988 population. In the 2SLS columns, exposure is instrumented with the one-year-lagged shift-share predictor described in Section 4. Columns (1)–(2) and (4)–(5) use 1992–2007. Columns (3) and (6) add 1985–1988, when measured exposure and the instrument are zero by construction, and exclude the transition years 1989–1991. Columns (2) and (5) include the baseline controls (Section 3.4) and the 1987 level of the outcome (the divorce rate in Panel A, the marriage rate in Panel B); columns (3) and (6) interact the baseline controls with year indicators. Standard errors clustered by district in parentheses. Anderson–Rubin 95% confidence sets, which are robust to weak instruments: for divorces, $[-0.055, -0.022]$ in column (5) and $[-0.050, -0.010]$ in column (6); for marriages, $[-0.109, -0.016]$ and $[-0.065, 0.018]$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

fewer divorces per 1,000 residents relative to a district without exposure, or a fifth of the sample mean. Because measured exposure begins in 1991 and omits the 1989–1990 surge, which flowed through the same networks, the estimates are effects per unit of measured post-1991 exposure and likely overstate the effect per actual arrival. In addition, as shown below, the decline is a post-reunification shift across districts rather than an effect that grows with each additional arrival.

Figure 3 reports the corresponding reduced-form event study for divorce: coefficients on the 1961 share interacted with year indicators, relative to 1988. Before reunification, the coefficients are small and jointly insignificant ($p = 0.26$), and the transition years 1989–1991 show no response. The coefficients turn negative from 1994: about −0.04 per percentage point of the share in 1994–1997, about −0.02 in 1998–2001, and between −0.06 and −0.07 in 2002–2005.

The decline increases with the historical share across the whole distribution. Relative to districts in the bottom quintile of the 1961 share, divorce fell after reunification by 0.09

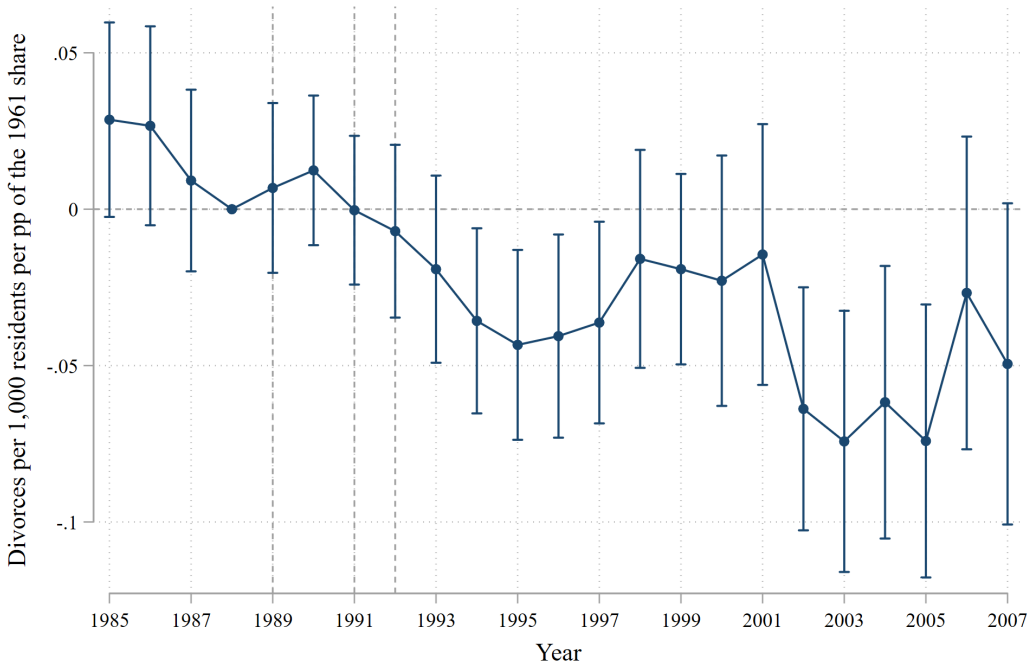


Figure 3: Event study of divorce by historical East-linked settlement share, 1985–2007

Notes: The figure plots the coefficients θ_k of equation (11) for divorces, with the share in percentage points; 320 districts, 1985–2007. Bars show 95% confidence intervals based on standard errors clustered by district. Vertical lines mark 1989, 1991, and 1992; 1989–1991 are transition years in which exposure is not measured.

per 1,000 residents in the second quintile, by 0.24 and 0.32 in the third and fourth, and by 0.35 in the top quintile, about 14 percent of the sample mean (Appendix Table B.1). Because these quintile comparisons do not rely on a linear functional form, they also show that the estimates are not driven by a few districts with extreme shares.

The timing indicates a shift rather than a cumulative effect. The decline emerges in the mid-1990s, after the largest inflow wave, and does not deepen in proportion to later arrivals: within districts over 1992–2007, the effect of exposure is small and imprecise (-0.015, s.e. 0.010; Appendix Table B.5), and in the lagged-outcome specification the coefficient per unit of exposure is largest in 1992–1996 (Appendix Table B.2). The IV coefficient therefore scales a post-reunification shift across districts by predicted exposure; it is not the marginal effect of an additional arrival within a district over time. The timing is consistent with a response to the inflows of the early 1990s, given that German divorce law generally requires a year of separation before a divorce can be granted.

Panel B of Table 5 shows no significant increase in marriages in any specification. With state-by-year fixed effects only and in the lagged-outcome specification, marriages fall, by 0.106 and 0.056 per arrival; the Anderson–Rubin set for the latter excludes zero. In the preferred specification, the estimate is -0.021, and its Anderson–Rubin set includes

zero.[4] The marriage event study, reported in Appendix B.2, shows no differential trend before reunification ($p = 0.52$) and negative coefficients afterward, but marriages do not decline monotonically across quintiles of the share (Appendix Table B.1). The marriage decline is therefore not robust; the robust finding is the absence of an increase.

5.2 Robustness

Table 6 reports variations of the preferred specification. Column (1) repeats the preferred estimate. Panel A changes the specification, and Panel B changes the sample, the weighting, and the inference.

Table 6: Robustness of the preferred estimate

Panel A. Specification				
	Preferred	Border-distance bins × year	Shift excludes own district only	Shift uses all districts
	(1)	(2)	(3)	(4)
Cumulative exposure $_{t-1}$	−0.0273***	−0.0399**	−0.0253***	−0.0249***
	(0.0094)	(0.0154)	(0.0077)	(0.0076)
Kleibergen–Paap F	27.40	12.17	36.48	37.17
Observations	6,400	6,400	6,400	6,400
Panel B. Sample, weighting, and inference				
	Trimmed exposure	Population weights	Regional clustering	Long difference
	(5)	(6)	(7)	(8)
Cumulative exposure $_{t-1}$	−0.0455***	−0.0413**	−0.0273***	−0.0245**
	(0.0154)	(0.0190)	(0.0101)	(0.0104)
Kleibergen–Paap F	20.53	12.65	18.25	26.24
Observations	5,740	6,400	6,400	320

Notes: The dependent variable is annual divorces per 1,000 residents of the fixed 1987 population. All columns except (8) use the preferred specification: district fixed effects, state-by-year fixed effects, and baseline controls interacted with year indicators, on 1985–1988 and 1992–2007. Column (2) adds fixed effects for 50-kilometer bins of distance to the former inner-German border interacted with year indicators. Columns (3) and (4) replace the instrument with versions whose shift excludes only the district's own arrivals or includes arrivals in all West German districts (Section 4). Column (5) excludes districts below the 5th and above the 95th percentile of cumulative exposure in 2007. Column (6) weights by 1987 population. Column (7) clusters standard errors by the 81 spatial planning regions (*Raumordnungsregionen*); all other columns cluster by district. Column (8) regresses the change in mean divorce rates between 1985–1987 and 2005–2007 on mean exposure in 2005–2007, instrumented with the mean instrument, with state fixed effects, baseline controls, and the 1987 divorce rate, and reports robust standard errors; the Anderson–Rubin test rejects zero at $p = 0.017$. Anderson–Rubin 95% confidence sets: $[-0.050, -0.010]$ in column (1) and $[-0.091, -0.014]$ in column (2). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[4] Four districts with exceptionally high marriage rates relative to their population—Nordfriesland, Friesland, and Wittmund, which include North Sea islands, and Baden-Baden—are popular wedding locations. Excluding them reduces the lagged-outcome estimate to −0.033 (s.e. 0.017) and the preferred estimate to −0.002 (s.e. 0.016).

Column (2) adds fixed effects for 50-kilometer bins of distance to the former inner-German border interacted with year indicators. These absorb shocks that vary with distance to the border, including changes in market access and the phase-out of place-based support for the former *Zonenrandgebiet* between 1991 and 1994, which coincides with the emergence of the divorce decline (Ehrlich & Seidel, 2018; Redding & Sturm, 2008). The estimated decline rises to 0.040. The first stage weakens ($F = 12.2$), but the Anderson–Rubin set still excludes zero. Columns (3) and (4) calculate the aggregate arrival rate from other sample districts and from the full sample, respectively; the estimates remain essentially unchanged (-0.025). Because these variants reweight a largely common aggregate path, they show that the results do not depend on how the shift is constructed, but they are not independent validation.

Panel B shows that the decline is not driven by districts with extreme exposure: excluding districts below the 5th and above the 95th percentile of cumulative exposure in 2007 yields -0.046 (column 5). Weighting districts by their 1987 population yields -0.041, with a weaker first stage (column 6). Clustering standard errors by the 81 spatial planning regions, which allows for correlation among neighboring districts, leaves the estimate statistically significant (column 7). Finally, a long-difference specification that compares mean divorce rates in 2005–2007 with those in 1985–1987 across districts yields -0.025 (column 8). Across specifications, the estimated decline ranges from 0.025 to 0.046 per arrival per 1,000 residents. The sign is stable, but the estimates should not be read as a precise structural parameter.

5.3 Pre-reunification trends and income growth

Under the identifying assumption (12), the coefficients θ_k in equation (11) are zero before reunification. The two long divorce panels described in Section 3.1 allow this to be tested over most of the 1980s; Table 7 reports the tests.

The longer pre-periods show no differential trends[5] In the districts observed since 1982 and since 1980, the pre-reunification interactions are jointly insignificant ($p = 0.34$ and $p = 0.16$), and the linear pre-trend in the share is close to zero. Over 1985–1988 alone, the interactions are also jointly insignificant, but the linear slope is negative and marginally significant (column 1); the longer windows show that this short-run movement is not part of a longer-run trend. In the same balanced districts, divorce declined after reunification by an average of 0.039 per percentage point of the share in the panel observed since 1982, which is statistically significant, and by a similar 0.033, less precisely estimated, in the smaller panel observed since 1980. Figure 4 plots the full event study for the districts observed since 1982, and Appendix Figure B.1 for those observed since 1980.

[5]With all district-years observed in 1980–1988, an unbalanced panel in which the set of states changes over the pre-period, the joint test rejects, but the linear pre-trend is close to zero (Appendix B.3).

Table 7: Pre-reunification trends in divorce

	All districts	Balanced since 1982	Balanced since 1980
	(1)	(2)	(3)
Joint pre-trend test (p)	0.261	0.341	0.156
Linear pre-trend	−0.0103*	−0.0012	0.0011
	(0.0053)	(0.0034)	(0.0043)
Mean 1992–2007 coefficient		−0.0390**	−0.0330
		(0.0181)	(0.0244)
Pre-period	1985–88	1982–88	1980–88
Districts	320	241	130
States	8	6	4

Notes: Each column estimates equation (11) for divorces, with the share in percentage points. Balanced panels contain the districts observed in every year from the first pre-period year to 2007. *Joint pre-trend test*: p-value for the hypothesis that all pre-period interactions are zero. *Linear pre-trend*: coefficient on the share multiplied by (year − 1988). Both are estimated on pre-period years only. *Mean 1992–2007 coefficient*: average of the 1992–2007 interactions from the full-window event study on the same districts. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

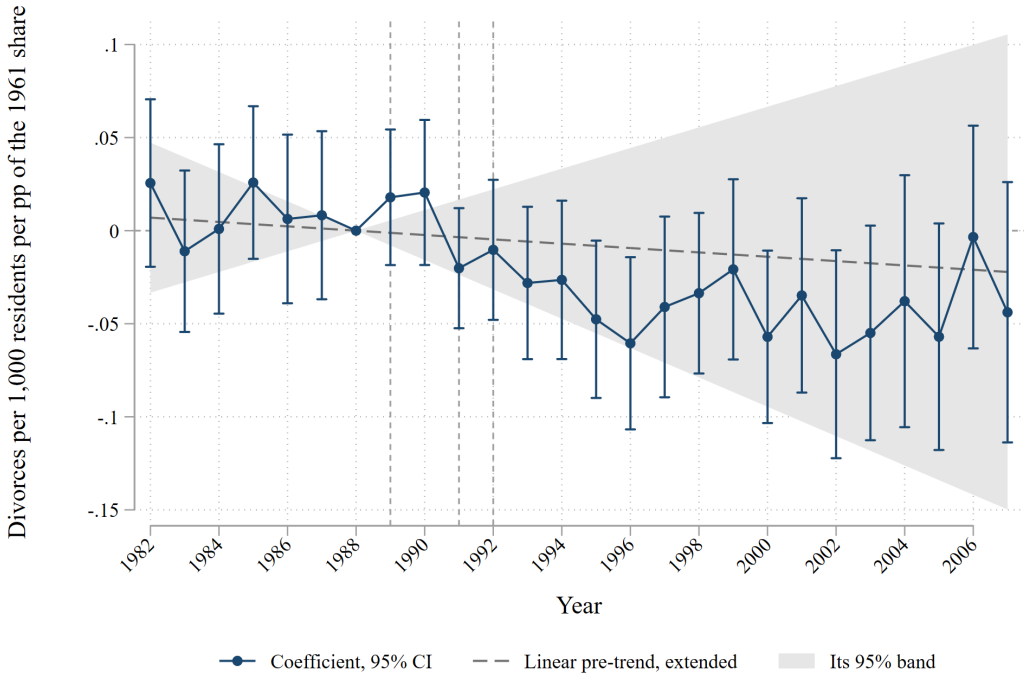


Figure 4: Event study of divorce for districts observed since 1982

Notes: The figure plots the coefficients θ_k of equation (11) for divorces, with the share in percentage points, for the 241 districts in six states observed in every year from 1982 to 2007; standard errors are clustered by district.

Two qualifications apply. First, a pre-period of seven to nine years cannot rule out slow-moving differential trends: a linear trend small enough to be consistent with the

pre-reunification data could, if it continued for two decades, account for much of the post-reunification decline.[6] Measuring the post-reunification coefficients relative to an extrapolated share-specific pre-trend leaves their average negative (−0.030 in the panel observed since 1982) but imprecise (s.e. 0.047). District-specific trends estimated on the post-reunification years are not a better test, because they absorb the response itself (Wolfers, 2006); with them, the estimate is close to zero and imprecise (0.001, s.e. 0.028). Second, in the lagged-outcome specification, districts with higher 1987 divorce rates experienced slower subsequent growth in divorce; allowing for this convergence reduces the lagged-outcome estimate from −0.036 to −0.023, which remains significant. Appendix Table B.4 reports all three trend adjustments.

The second threat in Section 4.3 includes an income channel: West German regions with more ties to the East grew faster after reunification (Burchardi & Hassan, 2013). Newly assembled district income data show that this channel is absent from the within-state variation that identifies the estimates. I estimate equation (11) with log income per inhabitant from the regional accounts as the outcome and 1989 as the reference year (Appendices A.4 and B.5). By 1995, the end of the period over which Burchardi and Hassan (2013) measure faster growth, income per inhabitant had not grown faster in districts with stronger ties in any specification (Table B.7, column 2); conditional on 1989 income, the difference is −0.1 percent per percentage point of the share (s.e. 0.2), whereas their estimates imply about 2.4 percent per percentage point of their tie measure. The differences in 1992 are small, at most 0.4 percent, and GDP per inhabitant and per person employed did not grow faster either. Consistent with this, the divorce estimate is robust to controlling for income: with year-specific effects of 1989 income it is −0.030 (s.e. 0.011), and with log income per inhabitant as a control −0.027 (s.e. 0.010), against −0.027 (s.e. 0.009) in the preferred specification, and the pre-reunification coefficients remain jointly insignificant ($p = 0.26$; Table B.8). The small remaining income differences could account for the estimated decline only if each percent of income per inhabitant changed the divorce rate by more than 5 percent; for elasticities between −1 and 1, the estimate lies between −0.031 and −0.023 (Conley et al., 2012).

These checks support the identifying assumption: districts with different historical East ties followed similar divorce paths before reunification, and the decline is explained neither by shocks that vary with distance to the former border nor by differences in income growth. The remaining assumption is that no slow-moving differential trend or other early-1990s shock correlated with historical East ties changed divorce.

[6] A differential trend of −0.0034 per percentage point of the share per year would generate the average post-reunification coefficient in column (2). It lies inside the confidence interval of the estimated pre-trend, and a test of no pre-trend has about 17 percent power against it (Roth, 2022).

5.4 Mechanisms

This subsection confronts the evidence with the predictions in Table 1. The marriage results already discriminate between the first and third hypotheses: a thicker partner market should raise both marriages and divorces, whereas higher costs of forming and dissolving households should lower both. Divorces fall and marriages do not rise in any specification, which is inconsistent with Hypothesis 1.

Table 8 uses the breakdown of divorces by filing party, available for Lower Saxony, North Rhine-Westphalia, and Baden-Württemberg, to assess Hypothesis 2. Under that hypothesis, a female-skewed inflow weakens wives' outside options and should reduce wife-initiated divorces more than husband-initiated ones. The estimates use the lagged-outcome specification, whose first stage remains strong in this smaller sample.

Table 8: Divorces by filing party

	Total	Husband-filed	Wife-filed	Joint	Wife minus husband
	(1)	(2)	(3)	(4)	(5)
Panel A. OLS					
Cumulative exposure $_{t-1}$	−0.0025	−0.0008	−0.0011	−0.0006	
	(0.0049)	(0.0023)	(0.0038)	(0.0029)	
Panel B. 2SLS					
Cumulative exposure $_{t-1}$	−0.0288***	−0.0010	−0.0110	−0.0168*	−0.0049
	(0.0107)	(0.0053)	(0.0083)	(0.0086)	(0.0089)
Kleibergen–Paap F	20.10	20.10	20.10	20.10	20.10
Observations	2,165	2,165	2,165	2,165	2,165
Districts	141	141	141	141	141

Notes: The sample consists of district-years in Lower Saxony, North Rhine-Westphalia, and Baden-Württemberg, 1992–2007, in which divorces are reported by filing party and the three components sum to the total; Lower Saxony does not report the breakdown in 1995 and 1997. Columns (1)–(4) use divorces per 1,000 residents of the fixed 1987 population; the components in columns (2)–(4) sum to column (1). Column (5) uses the change since 1987 in wife-filed minus husband-filed divorces per 1,000 residents. All regressions use the lagged-outcome specification: state-by-year fixed effects, baseline controls, and the 1987 divorce rate. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In the filing sample, total divorces fall by 0.029 per arrival, similar to the full-sample estimates. Husband-filed divorces do not change, wife-filed divorces fall by 0.011, and jointly filed divorces fall by 0.017, the largest component. The difference between wife- and husband-filed divorces is small and insignificant (−0.005, s.e. 0.009). These results do not support a one-sided shift in bargaining power. The concentration of the point estimates in jointly filed divorces is instead consistent with a change in the joint value of separation, as Hypothesis 3 implies, although none of the components is precisely estimated and the evidence should be read as exploratory.

The magnitude of the decline also argues against Hypothesis 2. Equation (3) implies that one additional young arrival per 1,000 residents lowers the local ratio of men to women aged 20–34 by about 0.0009. The preferred estimate corresponds to a decline in divorce of about 1.1 percent of the mean per arrival, and the lagged-outcome estimate to 1.4 percent. A sex-ratio channel would therefore require divorce to fall by roughly 14 to 18 percent for every one-point reduction in the number of men per 100 women, far beyond the district-level sex-ratio effects estimated for Germany, which are not robustly different from zero (Obersneider et al., 2019). Including the 1989–1990 surge, which was male-dominated, would move local sex ratios in the opposite direction.

Three further results narrow the interpretation. First, births do not respond beyond what the arrivals' own fertility could account for: the preferred estimate is 0.006 (s.e. 0.027), and in the lagged-outcome specification neither total births nor the split between marital and nonmarital births changes significantly (Appendix B.4). Second, the decline in divorce does not reflect a smaller stock of marriages at risk: it emerges in the mid-1990s, when that stock consisted overwhelmingly of marriages formed before reunification, and marriage formation does not fall robustly. Third, the district data cannot rule out that arrivals displaced incumbent residents; if young East Germans led some West German residents to move away, local marriage and divorce counts could fall without any change in behavior (Borjas, 2006).

Taken together, the evidence rejects Hypothesis 1 and does not support Hypothesis 2. It is consistent with Hypothesis 3, under which arrivals raised the costs of forming and dissolving households, and with the displacement of incumbent residents. The district data cannot separate these two explanations.

5.5 Heterogeneity

I next examine whether the response varies across local environments, by district type and by distance to the former inner-German border. Because the preferred specification has limited first-stage power in subsamples, the heterogeneity tables report OLS and 2SLS estimates with state-by-year fixed effects only and in the lagged-outcome specification. These estimates are descriptive, and whenever the Kleibergen–Paap statistic is low, the corresponding 2SLS coefficient is uninformative.

Table 9 distinguishes independent cities from counties. In counties, the lagged-outcome estimate is -0.019 with a strong first stage ($F = 37.7$). In independent cities, the estimate is close to zero, but the first stage is weak ($F = 11.2$), so the estimate is uninformative rather than evidence of no effect.

Table 10 splits districts by distance to the former inner-German border. Within 100 kilometers, the lagged-outcome estimate is -0.028 with an adequate first stage ($F = 17.1$). Farther from the border, the first stage is weak ($F = 3.7$ for 100–200

Table 9: Heterogeneity by district type

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Panel A. Independent cities (kreisfreie Städte)				
Cumulative exposure $_{t-1}$	−0.0064	−0.0076	−0.0015	−0.0054
	(0.0054)	(0.0048)	(0.0237)	(0.0187)
Kleibergen–Paap F			6.74	11.24
Observations	1,376	1,376	1,376	1,376
Panel B. Counties (Landkreise)				
Cumulative exposure $_{t-1}$	0.0061	0.0029	0.0437***	−0.0185**
	(0.0043)	(0.0035)	(0.0142)	(0.0078)
Kleibergen–Paap F			26.19	37.65
Observations	3,744	3,744	3,744	3,744
State × year fixed effects	✓	✓	✓	✓
Baseline controls		✓		✓
1987 divorce rate		✓		✓

Notes: The dependent variable is annual divorces per 1,000 residents of the fixed 1987 population, 1992–2007. Columns (3) and (4) instrument lagged cumulative exposure with the shift-share instrument. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

kilometers and 4.7 beyond 200 kilometers), and the estimates are uninformative. In both tables, the specifications with state-by-year fixed effects only again yield positive or imprecise coefficients, reflecting the persistent level differences discussed in Section 5.1. The response is thus precisely estimated in counties and near the former border, where migration costs were lower or partner markets thinner, but the weak first stages elsewhere do not allow the conclusion that the effect is absent there.

6 Conclusion

This paper studies how young East–West migration after German reunification affected marriage formation and marital dissolution in receiving West German districts. I combine a newly assembled panel of marriage, divorce and birth outcomes for 320 districts over 1985–2007 with observed cumulative arrivals aged 18–29 from 1991 onward. To address endogenous destination choice, I instrument exposure using 1961 East-linked settlement shares interacted with a cumulative young-arrival rate constructed outside each district's own state. The exposure measure includes arrivals of both sexes, and the estimates do not identify a female-specific effect.

The main empirical result is that districts with greater predicted exposure experienced a relative decline in divorce. In the preferred specification, which includes district and state-by-year fixed effects and year-specific effects of predetermined district charac-

Table 10: Heterogeneity by distance to the former inner-German border

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Panel A. Within 100 km				
Lagged cumulative exposure	0.0018	−0.0054	0.0080	−0.0281***
	(0.0045)	(0.0041)	(0.0090)	(0.0082)
Kleibergen–Paap F			8.81	17.09
Observations	1,360	1,360	1,360	1,360
Panel B. 100–200 km				
Lagged cumulative exposure	0.0047	0.0023	0.0592*	−0.2059*
	(0.0064)	(0.0063)	(0.0351)	(0.1141)
Kleibergen–Paap F			9.27	3.71
Observations	2,000	2,000	2,000	2,000
Panel C. Above 200 km				
Lagged cumulative exposure	0.0120	0.0024	0.1109**	−0.0041
	(0.0081)	(0.0073)	(0.0508)	(0.0367)
Kleibergen–Paap F			8.45	4.69
Observations	1,744	1,744	1,744	1,744
State × year fixed effects	✓	✓	✓	✓
Baseline controls		✓		✓
1987 divorce rate		✓		✓

Notes: The dependent variable is annual divorces per 1,000 residents of the fixed 1987 population, 1992–2007. Distance is measured from each district to the former inner-German border. Columns (3) and (4) instrument lagged cumulative exposure with the shift-share instrument. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

teristics, one additional cumulative young arrival per 1,000 residents of the fixed 1988 population corresponds to 0.027 fewer annual divorces per 1,000 residents of the fixed 1987 population. A one-standard-deviation increase in exposure corresponds to approximately 9 percent of the mean divorce outcome. The first stage is strong, and the weak-instrument-robust confidence set excludes zero. The negative estimate persists across alternative instrument constructions, border-distance controls, income controls, sample restrictions, population weighting and regional clustering.

The evidence for marriage formation is less conclusive. The preferred marriage estimate is negative, at −0.021, but statistically imprecise, and the estimated decline is sensitive to the specification and the inclusion of districts that attract weddings from elsewhere. The results therefore do not establish a robust decline in marriage formation. Supporting analyses of total births and births by parents' marital status likewise yield no statistically significant responses. These findings limit what can be inferred about a broader change in family formation.

The timing of the divorce response also shapes its interpretation. The decline emerges

in the mid-1990s and does not deepen in proportion to subsequent arrivals. The IV coefficient therefore primarily scales a post-reunification difference across districts by their predicted cumulative exposure. This pattern limits its interpretation as a constant marginal effect of each additional arrival. Moreover, district-level exposure begins in 1991 and omits the substantial inflows of 1989 and 1990. If those earlier arrivals followed the same historical networks and affected later outcomes, their effects may also contribute to the estimated response.

The heterogeneity results remain suggestive. Negative divorce estimates are more precisely estimated in counties and in districts within 100 kilometers of the former inner-German border under the lagged-outcome specification. Weaker first stages and imprecise estimates in other subsamples prevent firm conclusions about differences across local environments. These comparisons help describe the geographical pattern of the response, but do not independently identify its mechanism.

The mechanism evidence is similarly limited. The filing decomposition does not establish a statistically significant difference between wife- and husband-initiated divorce responses. Jointly filed divorces have the largest negative point estimate, although the component estimates are imprecise. The negative aggregate divorce response is difficult to reconcile with a simple account in which expanded partner availability increases marital dissolution, but the evidence does not isolate a sex-ratio or bargaining channel. Higher costs of separation and changes in the composition or location of the resident population remain possible explanations. Because the outcomes are event counts scaled by fixed baseline populations, the estimates cannot distinguish changes in couples' behavior from changes in the number and composition of couples residing in a district.

The main caveat remains identification. Balanced district panels extending back to 1980 and 1982 do not reject the null of no differential pre-reunification divorce trends. However, these tests have limited power against gradual divergence, and estimates measured relative to an extrapolated pre-reunification trend remain negative but imprecise. Border-distance-by-year fixed effects absorb shocks that vary with distance to the former border, and newly assembled district income data show that districts with stronger ties did not experience faster income growth after reunification; the estimate is robust to controlling for income. A causal interpretation therefore requires that, conditional on the fixed effects and predetermined characteristics, historical settlement ties affected subsequent family outcomes through migration exposure rather than through other channels.

The paper contributes a harmonized historical district panel and evidence linking reunification-era migration exposure to lower divorce activity in receiving West German districts. Under the identifying assumptions, the results indicate that the consequences of young internal migration extended to marital dissolution. Establishing whether this reflected greater stability within existing marriages, changes in the resident population, or adjustments in local economic conditions requires evidence beyond aggregate district

counts.

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A Data Construction

A.1 Divorce counts

The replication files list every corrected cell of the divorce panel, with the original transcription, the corrected count, the type of error and the source table. In a few state-years, the district counts could be compared only with the totals printed for administrative regions or for the state: Baden-Württemberg in 1981, 1982, 1984 and 2001, Lower Saxony in 2000 and 2001, and the Saarland in 1980–1984.

In four pairs of a city and its surrounding county, the published counts appear to assign divorces to the wrong member of the pair: Kassel in 2005–2007, where the Hessian statistical office notes the error, and Aachen, Oldenburg and Osnabrück in 2006–2007. These 18 district-years enter the analysis as published; dropping them changes the preferred estimate from −0.027 to −0.025 (s.e. 0.009).

The counts for the early 1980s contain abrupt changes in level. Of the 1,721 year-on-year changes observed for the 241 districts of the panel balanced since 1982, 73 exceed 40 percent in absolute value, 70 of them increases. Of these 73, 42 fall in five state-years: Bavaria in 1983 (13 districts), 1984 (6) and 1987 (10), North Rhine-Westphalia in 1981 (7), and Hesse in 1984 (6). The state-by-year fixed effects absorb the part of these changes that is common to the districts of a state. In 1985–1988, the counts used for the long pre-periods are identical to those of the main panel in all 1,280 district-years.

A.2 Migration counts

Apart from the merger that formed the Region Hannover in 2001, the West German districts do not change between 1991 and 2007, so the migration matrices need no boundary conversion for the 320 sample districts. Recomputing the 1991 state totals of young arrivals, and the 2001 total for Baden-Württemberg, directly from the original matrix files reproduces the processed counts exactly. Figure 2 plots these counts, summed over the 324 West German districts outside the city-states.

A.3 The 1961 East-linked share

The census district file reports R_i and $P_{i,1961}$ for the 565 districts of 1961, which the territorial reforms of the late 1960s and the 1970s merged and divided. I assign the counts of each 1961 district to the 1991 district or districts that contain its territory. R_i and $P_{i,1961}$ are converted in the same way, and s_i is computed from the converted counts.

Table A.1 sums the converted counts by state and compares them with the published census results. In the seven states whose districts are all in the sample, the sums equal the published counts up to the rounding of the published table, and the shares agree to the first decimal. Because any conversion that allocates every 1961 district in full preserves state totals, the comparison verifies the completeness of the census file and of the conversion, not the allocation of divided districts within a state. The Lower Saxony sums cover only the 43 sample districts and are therefore not comparable with the published total.

Table A.1: The 1961 East-linked population by state

	Published		Sample districts		
	Thousands	Share (%)	Number	Thousands	Share (%)
Schleswig-Holstein	114	4.9	15	114.3	4.9
Lower Saxony	356	5.4	43	252.0	4.7
North Rhine-Westphalia	909	5.7	54	908.8	5.7
Hesse	302	6.3	26	301.7	6.3
Rhineland-Palatinate	128	3.7	36	127.6	3.7
Baden-Württemberg	416	5.4	44	415.9	5.4
Bavaria	294	3.1	96	294.2	3.1
Saarland	23	2.1	6	22.8	2.1

Notes: Residents in 1961 who had lived in the Soviet zone or East Berlin in 1939 and moved to the Federal Republic after the war, with their children born since. Published: state results of the 1961 census reported by Ritter & Niehuss (1987, p. 31). Sample districts: sums of the converted counts R_i and $P_{i,1961}$ over the sample districts of each state. The Lower Saxony sample excludes the city and the district of Hannover, Göttingen and Osterode am Harz.

A.4 District income

District income comes from the regional accounts of the statistical offices. For 1989 and 1992, I digitized primary and disposable income of private households, as totals and per inhabitant, for all districts of the former West Germany from Arbeitskreis Volkswirtschaftliche Gesamtrechnungen der Länder (VGRdL, 1997), which reports both years in the vintage of autumn 1995. For 1995–2007, the same aggregates come from the August 2008 vintage (VGRdL, 2009b), and GDP and employment at the place of work for 1992 and 1994–2007 from its companion volume (VGRdL, 2009a). District income is not reported for 1990, 1991, 1993 and 1994. Deutsche Mark values are converted at 1.95583 per euro.

The digitized values pass three accounting checks: district totals add up to the printed totals of administrative regions and states; total and per-inhabitant income imply the same population in the tables for primary and disposable income, up to rounding; and the implied 1992 population matches that of the 2008 vintage within 0.004 percent in every district. One misread digit, in the 1989 total disposable income of Ulm, was corrected.

The two vintages follow different revisions of the national accounts. Between 1992 and 1995, the revision raised income per inhabitant in poorer rural districts relative to the richest ones: the dispersion of these changes across districts (a standard deviation of 0.057 log points) exceeds that of three-year changes within either vintage (0.040 and 0.029). Because districts with stronger historical ties had higher incomes in 1989—within states, one percentage point of the share goes with about 4.5 percent higher disposable income per inhabitant (s.e. 0.5)—comparisons across the two vintages are also reported with year-specific effects of log 1989 income, which absorb any revision that is linear in log income. Comparisons within a vintage need no such adjustment.

B Additional Results

B.1 Dose-response and timing

Table B.1 reports the comparisons across quintiles of the 1961 share, and Table B.2 the effect of exposure on divorce by period (Section 5.1).

Table B.1: Change after reunification by quintile of the 1961 share

	Divorces (1)	Marriages (2)
Second quintile × post	−0.0911**	0.1485
	(0.0432)	(0.1385)
Third quintile × post	−0.2449***	−0.0273
	(0.0600)	(0.1474)
Fourth quintile × post	−0.3160***	−0.1559
	(0.0650)	(0.1767)
Top quintile × post	−0.3532***	−0.1231
	(0.0683)	(0.1921)
Joint test, $F(4, 319)$ [p-value]	7.88 [<0.001]	1.03 [0.391]
Observations	6,400	6,400

Notes: Reduced-form regressions of the outcome on indicators for quintiles of the 1961 East-linked share, defined over all 320 districts, interacted with an indicator for 1992–2007; the bottom quintile is the reference. Fixed effects, controls and sample are those of the preferred specification: district fixed effects, state-by-year fixed effects, and baseline controls interacted with year indicators, on 1985–1988 and 1992–2007. Outcomes per 1,000 residents of 1987. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Effect of exposure on divorces by period

	1992–1996 (1)	1997–2001 (2)	2002–2007 (3)
Lagged cumulative exposure	−0.0684**	−0.0163	−0.0212**
	(0.0316)	(0.0111)	(0.0085)
Kleibergen–Paap F	15.78	23.22	28.01
Observations	1,600	1,600	1,920

Notes: 2SLS estimates in the lagged-outcome specification (state-by-year fixed effects, baseline controls, and the 1987 divorce rate), estimated separately for each period. Divorces per 1,000 residents of 1987. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Event-study coefficients

Table B.3 reports the coefficients of equation (11) for divorces, which Figure 3 plots, and for marriages.

Table B.3: Event-study coefficients for divorces and marriages

	Divorces		Marriages	
Year	Coefficient	Std. error	Coefficient	Std. error
1985	0.0286*	(0.0158)	−0.0066	(0.0231)
1986	0.0267	(0.0162)	0.0144	(0.0236)
1987	0.0092	(0.0148)	0.0105	(0.0170)
1988	Reference year			
1989	0.0068	(0.0138)	−0.0135	(0.0220)
1990	0.0124	(0.0122)	−0.0004	(0.0217)
1991	−0.0003	(0.0121)	0.0008	(0.0227)
1992	−0.0070	(0.0140)	−0.0236	(0.0258)
1993	−0.0192	(0.0152)	−0.0427	(0.0345)
1994	−0.0357**	(0.0150)	−0.0605*	(0.0318)
1995	−0.0434***	(0.0154)	−0.0716**	(0.0286)
1996	−0.0406**	(0.0165)	−0.0780**	(0.0344)
1997	−0.0362**	(0.0164)	−0.0805**	(0.0378)
1998	−0.0159	(0.0177)	−0.0853**	(0.0364)
1999	−0.0192	(0.0155)	−0.0631	(0.0386)
2000	−0.0229	(0.0203)	−0.0556	(0.0368)
2001	−0.0145	(0.0212)	−0.0847**	(0.0378)
2002	−0.0638***	(0.0197)	−0.0637*	(0.0375)
2003	−0.0742***	(0.0212)	−0.0827**	(0.0417)
2004	−0.0617***	(0.0221)	−0.0527	(0.0425)
2005	−0.0741***	(0.0222)	−0.0449	(0.0420)
2006	−0.0268	(0.0254)	−0.0388	(0.0434)
2007	−0.0495*	(0.0261)	−0.0656	(0.0417)
Joint test, 1985–1987: $F(3, 319)$	1.34 [$p = 0.261$]		0.75 [$p = 0.525$]	
Observations	7,360		7,360	

Notes: Coefficients θ_k of equation (11), with the share in percentage points: district fixed effects, state-by-year fixed effects, and baseline controls interacted with year indicators; 320 districts, 1985–2007, including the transition years 1989–1991. Outcomes per 1,000 residents of 1987. Standard errors clustered by district. For divorces, the joint test comes from the same specification estimated on 1985–1988, which yields identical coefficients for 1985–1987. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Pre-reunification trends

Figure B.1 shows the event study for the districts observed since 1980. Table B.4 reports the test on all district-years observed before reunification and the trend adjustments discussed in Section 5.3.

B.4 Births and alternative within-district specifications

Arrivals raise local birth counts at least through their own fertility, which gives a benchmark for the birth estimates. Attributing all births to residents aged 20–34, the sample means in Table 2 imply about $10.73/237 \approx 0.045$ births per resident of that age per year,

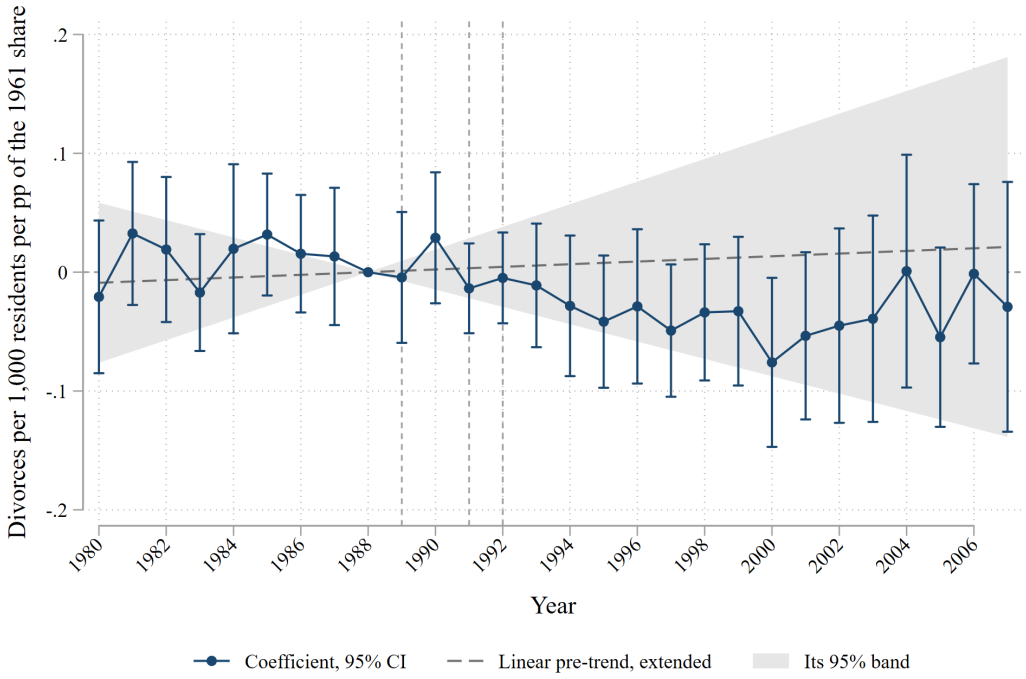


Figure B.1: Event study of divorce for districts observed since 1980

Notes: As Figure 4, for the 130 districts in North Rhine-Westphalia, Hesse, Baden-Württemberg and the Saarland observed in every year from 1980 to 2007. The dashed line extends the linear pre-trend estimated on 1980–1988, and the shaded area is its 95% confidence band.

where 237 is the number of residents aged 20–34 per 1,000 residents in 1987. If every measured arrival stayed and had this fertility, births per 1,000 residents would rise by about 0.045 per unit of exposure; return and onward moves, and the ageing of earlier arrivals, lower this benchmark further. An estimate far above it would indicate that the instrument picks up differences between districts other than the arrivals.

Table B.5 uses this benchmark to assess the specification. With district and state-by-year fixed effects alone (column 1), births rise by 0.178 per unit of exposure, four times the benchmark, and divorces fall by 0.054, twice the preferred estimate. Letting the baseline controls have year-specific effects (column 2) lowers the birth estimate to 0.056, and adding the pre-reunification years (column 3) lowers it to 0.006; neither differs significantly from the benchmark. The year-specific effects of the baseline controls therefore absorb differential trends that the share would otherwise pick up, which is why the preferred specification includes them and does not use the larger divorce estimate of column (1). Table B.6 reports births by the parents' marital status.

B.5 Income growth and the divorce estimate

Table B.7 reports equation (11) with log income per inhabitant as the outcome, estimated on the years in which district income is observed, with 1989 as the reference year; Figure B.2 plots the year-by-year coefficients of the second row of Panel A. Within each

Table B.4: Unbalanced pre-period test and trend adjustments, divorce

	Estimate	KP F	Districts	Observations
Panel A. All district-years observed in 1980–1988				
Joint pre-trend test (p)	0.021		320	2,278
Linear pre-trend	−0.0020		320	2,278
	(0.0029)			
Panel B. Mean 1992–2007 coefficient, net of the share-specific pre-trend				
Balanced since 1982	−0.0300		241	6,266
	(0.0465)			
Balanced since 1980	−0.0608		130	3,640
	(0.0591)			
Panel C. 2SLS with trend controls, 1992–2007				
District-specific trends	0.0013	49.88	320	5,120
	(0.0284)			
Lagged outcome with convergence	−0.0226***	27.10	320	5,120
	(0.0086)			
1987 divorce rate × years since 1991	−0.0250***			
	(0.0042)			
1985–1988 change × years since 1991	−0.0070			
	(0.0084)			

Notes: Divorces per 1,000 residents of 1987. KP F: Kleibergen–Paap first-stage F statistic. Panel A repeats the pre-period tests of Table 7 on all district-years observed in 1980–1988, an unbalanced panel in which states enter in different years (Section 3.1). Panel B replaces the pre-reunification share-by-year terms of the full-window event study with a linear trend in the share and averages the 1992–2007 coefficients, which then measure deviations from the extrapolated trend; Table 7 reports the unadjusted averages. In Panel C, the first row uses district fixed effects, district-specific linear trends and state-by-year fixed effects. The second row adds to the lagged-outcome specification the 1987 divorce rate and the change in the mean divorce rate from 1985–1986 to 1987–1988, each interacted with the number of years since 1991; both are measured per 1,000 residents. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

vintage, where no adjustment is needed, income in districts with stronger ties grew at essentially the same rate as elsewhere: by 0.22 percent more per percentage point of the share between 1989 and 1992 (s.e. 0.20; column 1, first row), and by 0.21 percent less on average after 1995 (s.e. 0.15; column 4). GDP at the place of work, available from 1992, did not grow faster either: relative to 1992, the mean coefficient over 1995–2007 is −0.0036 (s.e. 0.0040) for GDP per inhabitant and −0.0017 (s.e. 0.0029) for GDP per person employed.

Table B.8 shows that the divorce estimates do not depend on income. Adding year-specific effects of log 1989 income to the preferred specification raises the 2SLS estimate slightly, to −0.0298, leaves the event-study coefficients essentially unchanged, and keeps the pre-reunification coefficients jointly insignificant. Controlling for current log income per inhabitant, which may itself respond to the arrivals, leaves the 2SLS estimate at

Table B.5: Alternative within-district specifications

	District and state-year FE (1)	+ Baseline controls × year (2)	Preferred (3)
Divorces	−0.0537***	−0.0152	−0.0273***
	(0.0090)	(0.0103)	(0.0094)
Marriages	−0.0447**	0.0049	−0.0213
	(0.0173)	(0.0186)	(0.0195)
Live births	0.1780***	0.0556***	0.0060
	(0.0269)	(0.0195)	(0.0274)
Kleibergen–Paap F	57.69	29.01	27.40
Observations	5,120	5,120	6,400
District fixed effects	✓	✓	✓
State × year fixed effects	✓	✓	✓
Baseline controls × year		✓	✓
Pre-reunification years (1985–1988)			✓

Notes: 2SLS estimates of the effect of lagged cumulative exposure on outcomes per 1,000 residents of 1987. Columns (1) and (2) use 1992–2007. Column (3) is the preferred specification; its divorce and marriage estimates are those of Table 5. Within a column, all outcomes share the same first stage. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Births by the parents' marital status

	All births (1)	Marital births (2)	Nonmarital births (3)	Nonmarital share (%) (4)
Lagged cumulative exposure	0.0269	0.0353	−0.0084	0.0134
	(0.0308)	(0.0286)	(0.0060)	(0.0609)
Kleibergen–Paap F	38.29	38.29	38.29	38.65
Observations	5,120	5,120	5,120	5,120

Notes: 2SLS estimates in the lagged-outcome specification, 1992–2007: state-by-year fixed effects, baseline controls, and the 1987 levels of the birth rate and the nonmarital birth rate (columns 1–3) or of the nonmarital share (column 4). Columns (1)–(3) count live births per 1,000 residents of 1987; columns (2) and (3) sum to column (1). Column (4) is the percentage of live births to unmarried parents. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

−0.0271.

The bound in Section 5.3 follows Conley et al. (2012). If the share also affected divorce through income, with elasticity ε, the reduced form would contain the direct effect $\varepsilon\,\bar{y}\,\phi_Z$, and the 2SLS estimate net of it is

$$\beta(\varepsilon) = \frac{\rho - \varepsilon\,\bar{y}\,\phi_Z}{\pi}, \tag{13}$$

where $\rho = -0.0150$ and $\pi = 0.548$ are the reduced-form and first-stage coefficients of the preferred specification (Tables 4 and 3), $\bar{y} = 2.52$ is the mean divorce rate over 1992–

2007 (Table 2), and $\phi_Z = -0.0009$ (s.e. 0.0007) is the coefficient on Z_{it} in a regression of log disposable income per inhabitant on Z_{it}, on the years in which income is observed, with the fixed effects and controls of the preferred specification and year-specific effects of log 1989 income. The income channel would account for the entire estimate only at $\varepsilon = \rho/(\bar{y}\,\phi_Z) = 6.7$. The event-study analogue, the mean divorce coefficient over 1995–2007 in Table B.3 divided by $\bar{y}$ times the corresponding income coefficient (column 3 of Panel A, second row, of Table B.7), is 5.3.

Table B.7: Historical East ties and district income, 1989–2007

	Relative to 1989			Relative to 1995
	1992	1995	Mean 1995–2007	Mean 1996–2007
	(1)	(2)	(3)	(4)
Panel A. Log disposable income per inhabitant				
Preferred specification	0.0022	−0.0058**	−0.0077**	−0.0021
	(0.0020)	(0.0029)	(0.0031)	(0.0015)
+ 1989 income × year	0.0039**	−0.0011	−0.0031	
	(0.0017)	(0.0020)	(0.0021)	
+ border-distance bins × year	0.0027	−0.0024	−0.0037	
	(0.0019)	(0.0021)	(0.0023)	
Panel B. Log primary income per inhabitant				
Preferred specification	0.0002	−0.0071***	−0.0062***	0.0010
	(0.0016)	(0.0022)	(0.0023)	(0.0020)
+ 1989 income × year	0.0020	−0.0035	−0.0030	
	(0.0016)	(0.0027)	(0.0022)	

Notes: Coefficients of the 1961 East-linked share (in percentage points) interacted with year indicators, with district fixed effects, state-by-year fixed effects and baseline controls interacted with year indicators, as in the preferred specification; 320 districts in 1989, 1992 and 1995–2007 (4,800 observations). Columns (1) and (2) report the coefficients for 1992 and 1995 and column (3) the mean of the 1995–2007 coefficients. Column (4) reports the mean of the 1996–2007 coefficients from the same specification estimated on 1995–2007 alone (4,160 observations); within one vintage, no revision needs to be absorbed. Rows marked + add log disposable income per inhabitant in 1989 interacted with year indicators; the third row of Panel A also adds 50-kilometer bins of distance to the former inner-German border interacted with year indicators. A coefficient of 0.01 means 1 percent faster growth per percentage point of the share; the estimates of Burchardi & Hassan (2013) imply about 0.024 for 1989–1995. Data: Appendix A.4. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.8: Divorce estimates conditional on district income

	Preferred specification (1)	+ 1989 income × year (2)	+ Log income per inhabitant (3)
Panel A. 2SLS			
Cumulative exposure$_{t-1}$	−0.0273***	−0.0298***	−0.0271***
	(0.0094)	(0.0108)	(0.0097)
Kleibergen–Paap F	27.40	27.90	26.74
Observations	6,400	6,400	6,400
Panel B. Event study			
Mean 1995–2007 coefficient	−0.0418***	−0.0390***	
	(0.0136)	(0.0135)	
Joint pre-trend test, 1985–1987 (p)	0.261	0.258	
Observations	7,360	7,360	

Notes: The dependent variable is annual divorces per 1,000 residents of 1987. Column (1) repeats the preferred specification (Table 6, column 1, and Table B.3). Column (2) adds log disposable income per inhabitant in 1989 interacted with year indicators. Column (3) adds current log disposable income per inhabitant as a control, interpolated for 1993 and 1994 and set to its 1989 value for 1985–1988. Panel B estimates equation (11) and reports the mean of the 1995–2007 coefficients and the p-value of the joint test that the 1985–1987 coefficients are zero, estimated on 1985–1988 as in Table B.3. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

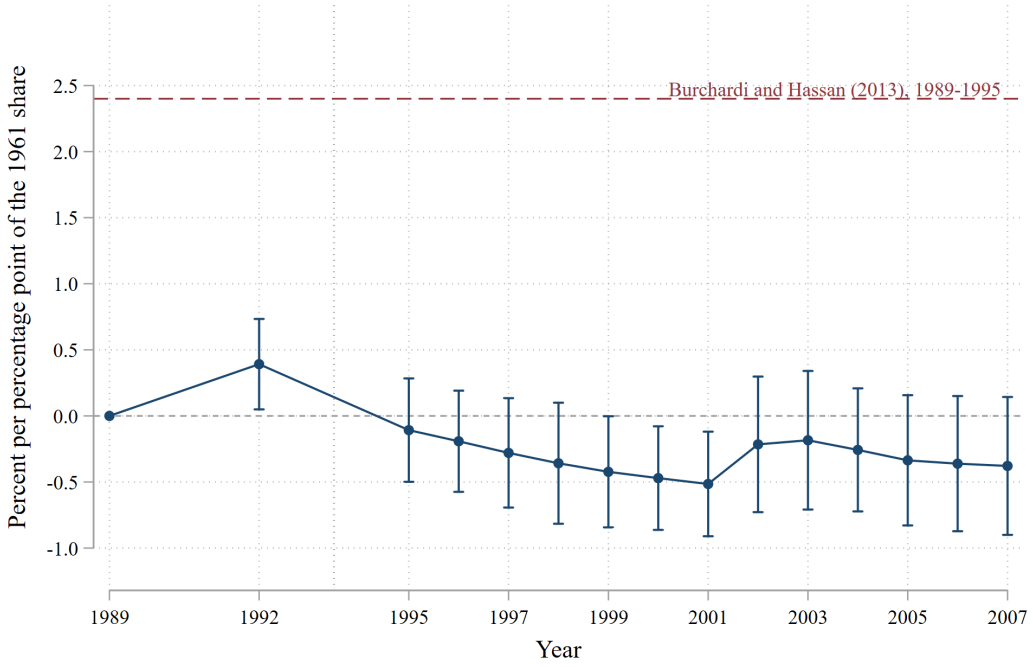


Figure B.2: Disposable income per inhabitant by historical East-linked share, 1989–2007

Notes: Coefficients of the 1961 share (in percentage points) interacted with year indicators, with log disposable income per inhabitant as the outcome and 1989 as the reference year, from the specification in the second row of Table B.7, Panel A, multiplied by 100. Bars show 95% confidence intervals based on standard errors clustered by district. The dotted line marks the change of vintage between 1992 and 1995; the dashed line marks the difference that the estimates of Burchardi & Hassan (2013) imply for 1989–1995, 2.4 percent per percentage point of their tie measure.